M13.4 Communication Batch 01 · Full Bank

Master-manual pages 1–117 · 400 questions · 180 from highlighted material (●).
Each topic: highlighted questions first, then the rest, each from Level 1 to Level 3. Answer key at the back, grouped by topic.

#	Topic	Pages	● High.	Total
1	Radio Technology Basics	6-7	8	15
2	AC Properties, Frequency and Angular Frequency	8-14	14	29
3	Harmonic, Triangular and Sawtooth Waves	15-20	5	13
4	Frequency Bands, Wavelength and Aviation Frequencies	21-31	24	48
5	Radio Wave Propagation: Ground Wave and Sky Wave	32-39	8	33
6	Ionosphere and Ionospheric Effects	40-47	22	50
7	Antenna Fundamentals, Fields, Polarization and Radiation Pattern	48-65	22	44
8	Antenna Types	66-82	26	52
9	Antenna Characteristics and Antenna Tuning	83-89	13	29
10	RF Lines, Matching and Standing Waves	90-103	26	45
11	Resonant Circuits	105-117	12	42

1 · Radio Technology Basics

Pages 6-7 · 15 questions

HIGHLIGHTED

1 **What do both transmitting and receiving equipment basically consist of?**

A Antennas, modulators and loudspeakers
B Electronic power supplies, amplifiers and oscillators
C Microphones, mixers and RF lines

Level 1 · Page 6

2 **What is the minimum a basic radio communication system may consist of?**

A A transmitter, a receiver and a relay
B An oscillator and an antenna
C A transmitter and a receiver

Level 1 · Page 6

3 **What is the transmitter stage that places the audio intelligence on the RF signal called?**

A The modulator
B The RF amplifier
C The oscillator

Level 1 · Page 6

4 **What converts sonic waves into electrical signals?**

A Loudspeakers
B Oscillators
C Microphones

Level 1 · Page 6

5 **What do loudspeakers convert?**

A Electrical signals into sonic waves
B Sonic waves into electrical signals
C Electrical signals into RF signals

Level 1 · Page 6

6 **How are the transmitter and the receiver of a basic radio system connected?**

A By the aircraft power supply lines
B By a matched RF line in all cases
C By the medium the waves travel through

Level 2 · Page 6

7 **Which statement about the propagation of electromagnetic waves is correct?**

A No medium is necessary for their propagation
B They can only travel through air
C They can only travel through a wire or a cable

Level 2 · Page 6

8 **How do the electromagnetic variations get from the transmitting antenna to the receiving antenna?**

A They are propagated through the medium (space)
B They are carried by the power supply lines
C They are conducted through the antenna feeder

Level 2 · Page 6

NOT HIGHLIGHTED

9 **What are the two broad categories of radio equipment?**

A Antenna and line equipment
B Transmitting and receiving equipment
C Analog and digital equipment

Level 1 · Page 6

10 **What does a transmitter comprise?**

A An oscillator, RF amplifiers and a modulator
B An amplifier, a detector and a loudspeaker
C A microphone, a loudspeaker and an antenna

Level 1 · Page 6

11 **What does the oscillator in a transmitter generate?**

A The modulated audio signal
B A basic radio frequency
C The audio intelligence

Level 1 · Page 6

12 **Which of these is named as a medium through which electromagnetic waves can travel?**

A A rubber insulator
B A dry wooden block
C A wave guide

Level 1 · Page 6

13 **Which unit feeds the transmitter in the basic radio system figure?**

A The microphone
B The receiver
C The loudspeaker

Level 1 · Page 7

14 **What does the receiving antenna convert the received electromagnetic energy into?**

A Sonic waves for the loudspeaker
B A steady direct current
C A flow of alternating RF currents

Level 2 · Page 6

15 **What does the receiver convert the RF current variations into?**

A A new basic radio frequency
B The intelligence in the transmission
C Electromagnetic waves for the antenna

Level 2 · Page 6

Coverage All text on pages 6–7 covered. Page 7 is a figure only.

2 · AC Properties, Frequency and Angular Frequency

Pages 8-14 · 29 questions

● **HIGHLIGHTED**

● **16** **Which types of current are used in communications engineering?**

A DC only
B Both AC and DC
C AC only

Level 1 · Page 8

● **17** **What is the time for one cycle to complete called?**

A The cycle time T
B The angular frequency ω
C The frequency f

Level 1 · Page 8

● **18** **What is the symbol for the period?**

A ω
B f
C T

Level 1 · Page 8

● **19** **What unit is the period measured in?**

A Radians per second
B Hertz (Hz)
C Seconds (s)

Level 1 · Page 8

● **20** **What is the frequency of AC?**

A The number of complete cycles in one second
B The number of half-waves in one second
C The time taken for one complete cycle

Level 1 · Page 10

● **21** **What does 1 cps equal?**

A 1 s
B 1 Hz
C 1 kHz

Level 1 · Page 10

● **22** **What is the formula for angular frequency?**

A $\omega = 2\pi \times f$
B $\omega = f / 2\pi$
C $\omega = 2\pi / f$

Level 1 · Page 12

23 **What is characteristic of AC?**

A The current varies in direction but not quantity
B The current varies in direction and in quantity
C The current varies in quantity but not direction

Level 2 · Page 8

24 **How does the polarity and current magnitude change of AC continue?**

A It happens only once at switch-on
B It repeats periodically
C It changes randomly with the load

Level 2 · Page 8

25 **What does one complete AC cycle consist of?**

A One positive alteration only
B Two positive alterations
C One positive and one negative alteration

Level 2 · Page 9

26 **How is frequency related to the cycle time T?**

A It is equal to T
B It is twice T
C It is the reciprocal of T

Level 2 · Page 10

27 **What does the rotational speed of the tip of the conductor or the magnet depend on?**

A The diameter of the generator only
B The RPM and the diameter of the generator
C The voltage and the current

Level 2 · Page 12

28 **How is a sinusoidal voltage represented as a phasor?**

A Length by the amplitude, rotation by the frequency
B Length by the cycle time, rotation by the phase
C Length by the frequency, rotation by the amplitude

Level 2 · Page 12

29 **How does angular velocity differ from angular frequency?**

A Both of them are vector quantities
B Velocity is a vector, frequency a scalar
C Velocity is a scalar, frequency a vector

Level 2 · Page 13

NOT HIGHLIGHTED

30 **Where is AC used in power engineering?**

A In battery circuits only
B In the mains supply
C In antenna feeders only

Level 1 · Page 8

31 **What is the abbreviation for alternating current?**

A AC
B RF
C DC

Level 1 · Page 8

32 **What waveform does every generator containing rotating parts create?**

A A sawtooth AC
B A rectangular AC
C A sinusoidal AC

Level 1 · Page 8

33 **What is the standard waveform for AC?**

A Rectangular
B Triangular
C Sinusoidal

Level 1 · Page 8

34 **What is the symbol for frequency?**

A T
B f
C ω

Level 1 · Page 10

35 **What unit is frequency measured in?**

A Seconds (s)
B Radians (rad)
C Hertz (Hz)

Level 1 · Page 10

36 **Which units is frequency very often expressed in?**

A kHz, MHz or GHz
B ms, µs or ns
C dB, dBi or dBm

Level 1 · Page 10

37 **How is a sinus typically created?**

A By a straight-line movement
B By a chemical reaction
C By a rotational movement

Level 1 · Page 12

38 **Which symbol represents both angular velocity and angular frequency?**

A T
B ω
C f

Level 1 · Page 13

39 **How does the voltage behave during the positive half-wave?**

A It rises from 0 to +peak and falls back to 0
B It rises from −peak to +peak
C It stays at the positive peak value

Level 2 · Page 8

40 **How does the voltage behave during the negative half-wave?**

A It rises from −peak to +peak
B It rises from 0 to −peak and falls back to 0
C It stays at the 0-reference line

Level 2 · Page 8

41 **What happens to the cycle time when the frequency increases?**

A It becomes longer
B It stays the same
C It becomes shorter

Level 2 · Page 10

42 **What is the unit of frequency expressed in seconds?**

A s^{-1}
B s
C s^{2}

Level 2 · Page 10

43 **What does the rotational speed depend on when the length is 1?**

A The diameter only
B The RPM and the voltage
C The RPM only

Level 2 · Page 12

44 **How do the formulas for angular velocity and angular frequency compare?**

A They are exactly the same
B Only one of them uses 2π
C They are reciprocals

Level 2 · Page 13

Coverage All text on pages 8–14 covered. Pages 11 and 14 are figures only. Page 9 figure labels used.
Conflict Page 9 (figure) uses the word 'Alteration' for each half of the cycle. Questions keep the LTT word.
Conflict Pages 12 and 13 both carry the LTT footer 'Page 8'; page 13 is an added page.

3 · Harmonic, Triangular and Sawtooth Waves

Pages 15-20 · 13 questions

HIGHLIGHTED

45 **What are multiples of the basic frequency called?**

A Sawtooth waves
B Further harmonic waves
C Basic oscillations

Level 1 · Page 16

46 **What is the sinusoidal wave with the same frequency as a non-sinusoidal wave called?**

A The 3rd harmonic or 2nd overtone
B The 2nd harmonic or 1st overtone
C The basic oscillation or 1st harmonic

Level 2 · Page 16

47 **What waveform results from superimposing the 1st, 3rd and 5th harmonics?**

A An almost sawtooth waveform
B An almost rectangular waveform
C An almost triangular waveform

Level 3 · Page 16

48 **Which harmonics are used to build a triangular wave?**

A The odd harmonics
B Both odd and even harmonics
C The even harmonics only

Level 3 · Page 19

49 **Which harmonics are required for a sawtooth wave?**

A The even harmonics only
B All harmonics, odd and even
C The odd harmonics only

Level 3 · Page 19

NOT HIGHLIGHTED

50 **Which other non-sinusoidal waves are possible besides the rectangular wave?**

A Triangular and sawtooth waves
B Sine and cosine waves
C Phasor and vector waves

Level 1 · Page 19

51 **How can non-sinusoidal oscillations be resolved to sine shape?**

A By mathematics or graphic means
B By raising the frequency
C By filtering with a resistor

Level 2 · Page 16

52 **Which harmonics can a rectangular wave be resolved into?**

A The 1st and 2nd harmonics only
B A finite number of even harmonics
C An infinite number of uneven harmonics

Level 2 · Page 16

53 **What determines the harmonic numbers and amplitudes?**

A The amplitude of the supply voltage
B The shape of the non-sinusoidal oscillation
C The length of the antenna

Level 2 · Page 16

54 **What would summing an infinite number of odd harmonics produce?**

A A triangular wave
B A square wave
C A sine wave

Level 2 · Page 18

55 **How is the amplitude of a harmonic of a rectangular wave calculated?**

A 1st harmonic amplitude ÷ harmonic number squared
B 1st harmonic amplitude × harmonic number
C 1st harmonic amplitude ÷ harmonic number

Level 3 · Page 16

56 **How does the harmonic amplitude decrease for a triangular wave?**

A In direct proportion to the harmonic number
B With the cube of the harmonic number
C With the square of the harmonic number

Level 3 · Page 19

57 **How does the harmonic amplitude decrease for a sawtooth wave?**

A In direct proportion to the harmonic number
B With the square of the harmonic number
C It does not decrease at all

Level 3 · Page 19

Coverage All text covered. Page 15 repeats page 16 (same text), cited as page 16. Pages 17, 18 and 20 are figures; page 18 caption used.

4 · Frequency Bands, Wavelength and Aviation Frequencies

Pages 21-31 · 48 questions

HIGHLIGHTED

58 **Which are the three most important frequency bands in communication engineering?**

A VLF, LF and MF
B HF, VHF and UHF
C LF, MF and HF

Level 1 · Page 21

59 **What is the frequency range of HF?**

A 3 MHz to 30 MHz
B 300 kHz to 3 MHz
C 30 MHz to 300 MHz

Level 1 · Page 21

60 **What is the frequency range of VHF?**

A 3 MHz to 30 MHz
B 300 MHz to 3000 MHz
C 30 MHz to 300 MHz

Level 1 · Page 21

61 **What is the frequency range of UHF?**

A 3 GHz to 30 GHz
B 300 MHz to 3000 MHz
C 30 MHz to 300 MHz

Level 1 · Page 21

62 **Which applications are grouped under communication in the frequency band chart?**

A Data communication, radio and satellite broadcast
B Inductive heating, annealing and melting
C Electro-medicine, drying and heating

Level 1 · Page 22

63 **What speed does a wave propagate with in free space (vacuum)?**

A The speed of sound
B Half the speed of light
C The speed of light c

Level 1 · Page 23

64 **How is the wavelength λ calculated?**

A $\lambda = c \times T = c / f$
B $\lambda = c \times f$
C $\lambda = f / c$

Level 1 · Page 23

65 **What is the approximate speed of light in a vacuum?**

A 300 000 km per second
B 30 000 km per second
C 3 000 000 km per second

Level 1 · Page 23

66 **How are radio frequencies up to 300 GHz divided?**

A Into octave ranges
B Into equal 1 GHz ranges
C Into decadic ranges

Level 1 · Page 27

67 **What is each decade of radio frequencies designated as?**

A A frequency band
B A channel
C A radar band

Level 1 · Page 27

68 **How are the frequencies used for radar technology divided?**

A Into audio frequency bands
B Into special radar bands
C Into decadic bands only

Level 1 · Page 27

69 **What is the frequency range of VHF COM?**

A 108 MHz to 118 MHz
B 118 MHz to 137 MHz
C 225 MHz to 400 MHz

Level 1 · Page 28

70 **What is the frequency range of HF COM?**

A 118 MHz to 137 MHz
B 3 kHz to 30 kHz
C 2 MHz to 30 MHz

Level 1 · Page 28

71 **What is the frequency range of UHF COM?**

A 962 MHz to 1213 MHz
B 225 MHz to 400 MHz
C 118 MHz to 137 MHz

Level 1 · Page 28

72 **What is the frequency range of the ADF?**

A 200 kHz to 1750 kHz
B 2 MHz to 30 MHz
C 10 kHz to 24 kHz

Level 1 · Page 28

73 **What frequency is shown for SatCom?**

A 4.3 GHz
B 9.375 GHz
C 1.5 GHz

Level 1 · Page 28

74 **What frequency is shown for weather radar?**

A 4.3 GHz
B 9.375 GHz
C 8.8 GHz

Level 1 · Page 28

75 **What frequency range is shown for VLF navigation?**

A 10 kHz to 24 kHz
B 200 kHz to 1750 kHz
C 300 Hz to 3 kHz

Level 1 · Page 28

76 **What is the identification signal frequency of the DME?**

A 1350 Hz
B 400 Hz
C 1020 Hz

Level 1 · Page 28

77 **What is the identification signal frequency of VOR, ILS and ADF?**

A 3000 Hz
B 1020 Hz
C 1350 Hz

Level 1 · Page 28

78 **What frequency range is shown for voice?**

A 3 kHz to 30 kHz
B 20 Hz to 20 kHz
C 300 Hz to 3 kHz

Level 1 · Page 28

79 **Which tone frequencies are shown for the marker?**

A 300, 1020 and 1350 Hz
B 1020, 1350 and 3000 Hz
C 400, 1300 and 3000 Hz

Level 1 · Page 28

80 **What does the propagation speed of a wave depend on?**

A The amplitude of the voltage
B The medium used
C The length of the antenna

Level 2 · Page 23

81 **What happens to the propagation speed as the density of the medium increases?**

A It stays the same
B It increases
C It decreases

Level 2 · Page 23

NOT HIGHLIGHTED

82 **What does RF stand for?**

A Reference Frequency
B Radio Frequency (carrier wave)
C Relative Frequency

Level 1 · Page 27

83 **What does IF stand for?**

A Intermediate Frequency
B Internal Frequency
C Isotropic Frequency

Level 1 · Page 27

84 **What does AF stand for?**

A Amplified Frequency
B Antenna Frequency
C Audio Frequency

Level 1 · Page 27

85 **Which radar band covers 5200 MHz to 10 900 MHz?**

A C band
B X band
C S band

Level 1 · Page 27

86 **What is the frequency range of the L band?**

A 1550 MHz to 5200 MHz
B 3900 MHz to 6200 MHz
C 390 MHz to 1550 MHz

Level 1 · Page 27

87 **What is the frequency range of the S band?**

A 1550 MHz to 5200 MHz
B 390 MHz to 1550 MHz
C 5200 MHz to 10 900 MHz

Level 1 · Page 27

88 **What is the frequency range of the C band?**

A 3900 MHz to 6200 MHz
B 1550 MHz to 5200 MHz
C 10 900 MHz to 36 000 MHz

Level 1 · Page 27

89 **Which radar band covers 10 900 MHz to 36 000 MHz?**

A C band
B X band
C K band

Level 1 · Page 27

90 **Which wavelength does 300 MHz match in the aviation frequencies chart?**

A 10 cm
B 10 m
C 1 m

Level 1 · Page 28

91 **What is the frequency range of the localizer?**

A 108 MHz to 112 MHz
B 329.3 MHz to 399.95 MHz
C 112 MHz to 118 MHz

Level 1 · Page 28

92 **What frequency range is shown for the VOR?**

A 112 MHz to 118 MHz
B 118 MHz to 137 MHz
C 108 MHz to 112 MHz

Level 1 · Page 28

93 **What is the frequency range of the glide slope?**

A 962 MHz to 1213 MHz
B 329.3 MHz to 399.95 MHz
C 108 MHz to 112 MHz

Level 1 · Page 28

94 **What frequency does the marker beacon use?**

A 75 MHz
B 108 MHz
C 440 MHz

Level 1 · Page 28

95 **What is the frequency range of the DME?**

A 329.3 MHz to 399.95 MHz
B 962 MHz to 1213 MHz
C 1030 MHz to 1090 MHz

Level 1 · Page 28

96 **Which frequencies does ATC use?**

A 962 MHz to 1213 MHz
B 4.25 GHz to 4.35 GHz
C 1030 MHz to 1090 MHz

Level 1 · Page 28

97 **What frequency range is shown for the LRRA?**

A 1.2 GHz to 1.5 GHz
B 8.8 GHz to 9.375 GHz
C 4.25 GHz to 4.35 GHz

Level 1 · Page 28

98 **What frequency is shown for the Doppler radar?**

A 8.8 GHz
B 9.375 GHz
C 4.3 GHz

Level 1 · Page 28

99 **Which frequencies are shown for GPS L1 and L2?**

A L1 1.2 GHz and L2 1.5 GHz
B L1 1.5 GHz and L2 1.2 GHz
C L1 4.3 GHz and L2 1.5 GHz

Level 1 · Page 28

100 **Which propagation type is listed for HF in the frequency band table?**

A Space wave
B Ground wave
C Sky wave

Level 1 · Page 30

101 **Which propagation type is listed for VHF and UHF?**

A Ground wave
B Space wave
C Sky wave

Level 1 · Page 30

102 **What general rule of thumb gives the upper limit of each band?**

A Multiply the lower limit by 10
B Multiply the lower limit by 2
C Add 30 MHz to the lower limit

Level 2 · Page 21

103 **What wavelength range belongs to the X band?**

A 77 cm to 20 cm
B 5.8 cm to 2.5 cm
C 20 cm to 5.8 cm

Level 2 · Page 27

104 **Which wavelength range matches the HF band?**

A 100 m to 10 m
B 10 m to 1 m
C 1 km to 100 m

Level 2 · Page 28

105 **Which band is listed as using both ground wave and sky wave?**

A MF
B HF
C VHF

Level 2 · Page 30

Coverage All text covered. Page 24 repeats page 23. Page 26 repeats the page 22 chart. Page 29 repeats the page 28 aviation frequencies as a picture. Page 30 table used for propagation types.
Conflict Page 27 radar band table: the C band (3900–6200 MHz) overlaps the S band (1550–5200 MHz) and the X band (5200–10 900 MHz). Questions use the table values as printed.
Conflict Page 28 (figure): 'ALTM 440 MHz' is highlighted but not explained anywhere; no question was written on it.
Conflict Page 31 is a general spectrum picture with no LTT text; not used for questions.

5 · Radio Wave Propagation: Ground Wave and Sky Wave

Pages 32-39 · 33 questions

● **HIGHLIGHTED**

● **106** **What speed do radio waves travel at?**

A Approximately the speed of sound
B Approximately the speed of light
C Exactly half the speed of light

Level 1 · Page 32

● **107** **What is the part of the radio wave that moves outward in contact with the ground called?**

A The sky wave
B The ground wave
C The space wave

Level 1 · Page 32

● **108** **What is the part of the radio wave that moves upward and outward called?**

A The ground wave
B The surface wave
C The sky wave

Level 1 · Page 32

● **109** **What does wave propagation define?**

A The way waves are transmitted from one end to another
B The rate at which a wave loses energy
C The direction of the electrical field

Level 1 · Page 34

● **110** **Which wave is responsible for most of the daytime broadcast reception?**

A The space wave
B The sky wave
C The ground wave

Level 1 · Page 36

● **111** **What is the sky wave used for in daylight?**

A Line-of-sight UHF communication
B Short-range low frequency communication
C Long-range high frequency communication

Level 2 · Page 32

● **112** **Which frequencies only travel in a direct line?**

A From HF and above
B MF and below
C LF and VLF only

Level 2 · Page 34

113 **What is the ground wave used for?**

A Long-range HF daylight communication only
B Short range at LF with low power, long range at HF with very high power
C Short range at HF with low power, long range at LF with very high power

Level 3 · Page 32

NOT HIGHLIGHTED

114 **What are radio waves propagated into space considered to be?**

A A direct current flowing through space
B A radiant form of energy like light and heat
C A mechanical vibration of the air

Level 1 · Page 32

115 **Which frequencies travel as space waves in a line of sight?**

A MF and lower
B HF only
C VHF and higher

Level 1 · Page 34

116 **Which frequencies follow the curvature of the earth as ground waves?**

A MF and lower
B VHF and higher
C UHF only

Level 1 · Page 34

117 **What does the term attenuation mean?**

A Bend or refract
B Reduce or weaken
C Increase or amplify

Level 1 · Page 36

118 **How much greater is the conductivity of salt water than that of dry soil?**

A 5000 times
B 500 times
C 50 000 times

Level 1 · Page 36

119 **What is the usual upper limit of the ground wave frequency range?**

A About 300 MHz
B About 30 MHz
C About 2 MHz

Level 1 · Page 37

120 **What is the permissible frequency range for sky wave propagation?**

A 300 kHz to 3 MHz
B 3 MHz to 30 MHz
C 30 MHz to 300 MHz

Level 1 · Page 39

121 **What does the field pattern of a wave leaving a vertical antenna resemble?**

A A narrow pencil beam
B A huge doughnut lying on the ground
C A figure eight standing upright

Level 2 · Page 32

122 **How is daytime reception from most commercial stations carried?**

A By sporadic-E ionisation
B By the sky wave
C By the ground wave

Level 2 · Page 32

123 **What does the sky wave provide at night?**

A Line-of-sight contacts at UHF
B Long-range contacts at somewhat lower frequencies
C Short-range contacts at higher frequencies

Level 2 · Page 32

124 **How does the ground wave lose energy as it travels?**

A It is refracted by the ionosphere
B It is reflected by the sea surface
C It induces eddy currents in the earth

Level 2 · Page 36

125 **What does increasing the frequency do to ground wave attenuation?**

A It rapidly increases it
B It has no effect on it
C It rapidly decreases it

Level 2 · Page 36

126 **Which frequencies do shore-based transmitters use for long-range ground wave transmission?**

A 3 MHz to 30 MHz with low power
B 18 kHz to 300 kHz with extremely high power
C 118 MHz to 137 MHz with low power

Level 2 · Page 36

127 **Why is the ground wave signal strength at a given point nearly constant?**

A The ionosphere does not change
B The transmitter power is regulated
C The earth's electrical properties are fairly constant

Level 2 · Page 36

128 **Where does the ground wave signal strength NOT stay nearly constant?**

A In areas close to the ocean
B In areas with distinct rainy and dry seasons
C In areas at high altitude

Level 2 · Page 36

129 **Why are high power, low frequency transmitters placed close to the ocean?**

A Salt water gives superior surface wave conduction
B Water reflects the sky wave upward
C Sea air reduces ionospheric absorption

Level 2 · Page 36

130 **How does the ground wave oscillate relative to the surface of the earth?**

A Parallel to the surface
B At right angles to the surface
C At the critical angle

Level 2 · Page 37

131 **What happens to some of the energy of the sky wave at the ionosphere?**

A It is turned into a ground wave
B It is fully absorbed
C It is refracted back toward the earth

Level 2 · Page 38

132 **How is some of the sky wave energy lost?**

A Eddy currents set up in the earth
B Dissipation to atmospheric particles
C Reflection from the sea surface

Level 2 · Page 38

133 **Which is an advantage of sky wave propagation?**

A It supports large distance propagation
B It needs only small antennas
C It is the same by day and by night

Level 2 · Page 39

134 **Which is a disadvantage of sky wave propagation?**

A Its atmospheric attenuation is high
B Long distances need large antennas
C It only covers short distances

Level 2 · Page 39

135 **Which applications widely use sky wave propagation?**

A Weather radar and radio altimeter
B Marker beacons and glide slope
C Mobile and satellite communications

Level 2 · Page 39

136 **What does the ground wave induce on the surface of the earth?**

A A wave of half size, same polarity
B A wave of double size, same polarity
C A wave of equal size, opposite polarity

Level 3 · Page 37

137 **Where can a receiver pick up strong sky wave signals?**

A Only inside the ground wave range
B Several hundred miles beyond the ground wave range
C Only directly below the transmitter

Level 3 · Page 38

138 **Why does sky wave transmission vary between day and night?**

A The ionosphere is nearer at night and farther by day
B The ground conductivity changes with daylight
C The transmitter power changes with daylight

Level 3 · Page 39

Coverage All text on pages 32–39 covered. Pages 33 and 35 are pictures only.
Conflict Page 34 says 'High frequencies (from HF and above) only travel in a direct line', but the same page and pages 32 and 39 say HF reflects from the ionosphere as a sky wave. Questions follow each statement as written.
Conflict Page 39 says sky wave propagation is 'widely used in mobile and satellite communications'. This does not fit the rest of the manual (satellite links are space waves). The question follows the LTT wording.
Conflict Page 39 says the ionosphere is 'near and far during night and day respectively'. The wording is unclear; the question follows it as written.

6 · Ionosphere and Ionospheric Effects

Pages 40-47 · 50 questions

● **HIGHLIGHTED**

● **139** **Where is the ionosphere found?**

A About 400 to 500 miles above the earth
B About 4 to 5 miles above the earth
C About 40 to 50 miles above the earth

Level 1 · Page 40

● **140** **Where does the F layer extend?**

A From 90 miles to the ionosphere's upper limit
B From 40 miles up to 50 miles only
C From 50 miles up to 90 miles only

Level 1 · Page 41

● **141** **What makes the ionosphere appear to have layers?**

A Temperature differences at different heights
B Densities of ionization at different heights
C The lines of the earth's magnetic field

Level 2 · Page 40

● **142** **What effect does the D layer have on radio waves?**

A Little effect, except absorption of energy
B Total reflection of VHF signals
C Strong refraction back to earth

Level 2 · Page 41

● **143** **When is the D layer present?**

A Both day and night
B Only during the day
C Only during the night

Level 2 · Page 41

● **144** **When does the maximum density of the E layer appear?**

A During early afternoon
B Shortly after sunset
C At about noon local time

Level 2 · Page 41

● **145** **When is the F2 layer at its greatest density as a rule?**

A Shortly after sunset
B At about noon local time
C During early afternoon hours

Level 2 · Page 41

146

What happens to the F1 and F2 layers shortly after sunset?

A They recombine into a single F layer
B They both disappear completely
C They separate into two parts

Level 2 · Page 41

147

What are erratic patches of ionized atmosphere at E layer heights called?

A Critical frequency zones
B D layer absorption
C Sporadic-E ionisation

Level 2 · Page 41

148

What can strong sporadic ionisations at varying altitudes do?

A Harm electronic transmissions
B Remove the D layer completely
C Double the critical frequency

Level 2 · Page 41

149

How do waves behave when they reach the ionosphere?

A All are reflected back to earth
B Some pass into space, others bend
C All are absorbed by the layers

Level 2 · Page 44

150

What is the angle made by the group 2 rays called?

A The skip angle
B The critical angle
C The elevation limit

Level 2 · Page 44

151

Which angle do the group 3 rays strike the ionosphere at?

A The smallest angle that still returns to earth
B The largest angle that still returns to earth
C An angle that is too nearly vertical

Level 2 · Page 44

152

What is the critical frequency?

A The frequency at which the D layer absorbs all energy
B The highest frequency sent straight up that still returns
C The lowest frequency sent straight up that still returns

Level 2 · Page 44

153

What does the skip distance measure in the skip zone figure?

A From the end of the ground wave to point B
B From the transmitter A to the first sky wave return B
C From the ionosphere down to point B

Level 2 · Page 46

154 **What daylight transmission distances does E layer refraction support?**

A Up to 1500 miles
B Up to 15 000 miles
C Up to 150 miles

Level 3 · Page 41

155 **What can sporadic-E ionisation enable?**

A Lower ground wave attenuation at LF
B Removal of the skip zone at night
C Good VHF transmissions over unusual distances

Level 3 · Page 41

156 **What does the ability of the ionosphere to return a wave to earth depend on?**

A The angle, the frequency and the ion density
B The power, the antenna height and the polarization
C The ground conductivity, the season and the antenna type

Level 3 · Page 44

157 **When will a wave emerge from the ionosphere and return to earth?**

A When the transmitter power is high enough
B When the wave is horizontally polarized
C When frequency, angle and ion density are suitable

Level 3 · Page 44

158 **Why are the group 1 rays not returned to earth?**

A Their frequency is too low
B Their angle is too nearly vertical
C Their angle is too close to the horizon

Level 3 · Page 44

159 **What happens to a ray that leaves the antenna at an angle greater than the critical angle?**

A It is absorbed by the D layer
B It penetrates the ionosphere
C It returns at a shorter distance

Level 3 · Page 44

160 **What happens to rays at an angle smaller than that of the group 3 rays?**

A They are refracted but do not return to earth
B They return to earth near the transmitter
C They pass straight through the ionosphere

Level 3 · Page 44

NOT HIGHLIGHTED

161 **What does the ionosphere contain more of than other parts of the atmosphere?**

A Ozone molecules
B Water vapour
C Positive and negative ions

Level 1 · Page 40

162 **Which altitude are air particles too sparse above for large-scale energy transfer?**

A 50 miles
B 350 miles
C 90 miles

Level 1 · Page 40

163 **What is the altitude of the D layer?**

A Between 40 and 50 miles
B Above 90 miles
C Between 50 and 90 miles

Level 1 · Page 41

164 **What altitude band contains the E layer?**

A Above 90 miles
B Between 40 and 50 miles
C Between 50 and 90 miles

Level 1 · Page 41

165 **Which altitude does the E layer have its greatest density at?**

A About 150 miles
B About 70 miles
C About 45 miles

Level 1 · Page 41

166 **What raises atoms in the ionosphere to a high energy level?**

A The earth's magnetic field
B Lightning in the lower atmosphere
C Solar ultraviolet and particle radiation

Level 2 · Page 40

167 **Which factors affect the number of ions in the ionosphere?**

A Earth's rotation, its course round the sun and sunspots
B Transmitter power, antenna height and polarization
C Ground conductivity, rainfall and sea state

Level 2 · Page 40

168 **What does the rate of variation between high and low energy levels depend on?**

A Amount of air, solar radiation strength and the earth's magnetic field
B Transmitter power, frequency and antenna type
C Ground conductivity, humidity and temperature

Level 2 · Page 40

169 **Why are there only a few ions below about 40 miles altitude?**

A The ultraviolet radiation is strongest there
B The rate of return to a normal energy level is high
C The air particles are far too sparse

Level 2 · Page 40

170 **How sharp are the dividing lines between the ionosphere layers?**

A There is thought to be no sharp line
B They are sharp only at night
C They are very sharp at all times

Level 2 · Page 40

171 **How does the D layer affect transmissions through daylight zones?**

A It greatly increases their range
B It changes their polarization
C It greatly reduces their field intensity

Level 2 · Page 41

172 **When is the E layer strongest?**

A During daylight hours
B Shortly after sunset
C During the night

Level 2 · Page 41

173 **How does the F layer change between night and day?**

A It is present only by day
B Two parts at night, one layer by day
C One layer at night, two parts (F1, F2) by day

Level 2 · Page 41

174 **Which layers remain at night in the ionosphere figure?**

A D and E
B F and E
C F1 and F2

Level 2 · Page 43

175 **What does the ionosphere do as an electronic mirror?**

A It absorbs the whole ground wave
B It changes the wave polarization
C It refracts the sky wave back to earth

Level 2 · Page 44

176 **What does the critical frequency vary with?**

A Ground conductivity and rainfall
B Locality, time, season and sunspots
C Antenna type, polarization, power

Level 2 · Page 44

177 **What do nomograms and frequency tables predict?**

A The ground wave range for each transmitter
B The Maximum Usable Frequency for each hour and locality
C The skip distance for each antenna type

Level 2 · Page 45

178 **How are the nomograms and frequency tables prepared?**

A From measured data from stations worldwide
B From the daily sunspot count alone
C From calculations of antenna gain

Level 2 · Page 45

179 **What is the skip zone?**

A The area covered by the ground wave alone
B The area beyond the first returning sky wave
C The silent area between ground wave end and first sky wave

Level 2 · Page 45

180 **What is multiple-hop transmission?**

A Transmission on several frequencies at once
B Reception of the ground and sky wave together
C Repeated reflection between ionosphere and ground

Level 2 · Page 45

181 **Which frequencies can the E layer refract back to earth in the middle of the day?**

A Up to 200 MHz
B Up to 20 MHz
C Up to 2 MHz

Level 3 · Page 41

182 **What happens to the initial angle as the frequency increases?**

A It decreases
B It increases
C It stays the same

Level 3 · Page 44

183 **What happens to the wave at sufficiently high frequencies?**

A It is not returned at any angle
B It is returned only when sent straight up
C It is returned at any angle

Level 3 · Page 44

184 **When is the skip zone for the lower high frequencies (3 to 9 MHz) greater?**

A At noon
B At night
C In early afternoon

Level 3 · Page 45

185 **How can skip effects be minimized at night?**

A By using lower frequencies
B By increasing the antenna height
C By using higher frequencies

Level 3 · Page 45

186 **What happens to the skip distance as the frequency decreases?**

A It increases
B It stays the same
C It decreases

Level 3 · Page 45

187 **What can cause fading with multiple-hop transmission?**

A Signals with different reflections superposing with different phases
B The antenna changing its polarization
C The D layer absorbing the ground wave

Level 3 · Page 45

188 **What happens to the critical angle and refraction as frequency increases?**

A The angle rises, refraction falls
B Both of them decrease
C Both of them increase

Level 3 · Page 47

Coverage All text on pages 40–47 covered. Pages 42–43 figures and page 47 picture text used.
Conflict Page 46 (figure): 'Skip Distance' is labelled from A to B in the figure only; the text never defines it. The question is marked as coming from a figure.

7 · Antenna Fundamentals, Fields, Polarization and Radiation Pattern

Pages 48-65 · 44 questions

● **HIGHLIGHTED**

● **189** **What does an antenna represent in a radio system?**

A The modulator of the transmitter
B The power supply of the transmitter
C The link between transmitter and receiver

Level 1 · Page 48

● **190** **What are antennas always supplied with?**

A DC from the aircraft bus
B AC, typically at high frequencies
C AC at mains frequency

Level 1 · Page 52

● **191** **What is the typical length of an antenna?**

A 2λ
B λ
C λ/4

Level 1 · Page 52

● **192** **What angle is there between the electrical and the magnetic field?**

A 0 degrees
B 90 degrees
C 180 degrees

Level 1 · Page 60

● **193** **What are the two most common types of antenna polarization?**

A Horizontal and vertical
B Linear and circular
C Circular and elliptical

Level 1 · Page 60

● **194** **What is an antenna?**

A An amplifier for the received RF signal
B A filter that selects one frequency
C A wave transformer between line energy and free radiation

Level 2 · Page 48

● **195** **How is the current distributed along a wire supplied with voltage?**

A Greatest at the feed point, zero at the end
B Equal along the whole wire
C Zero at the feed point, greatest at the end

Level 2 · Page 50

196

What shape does the current distribution on the wire have?

A Part of a sine
B A straight line
C A rectangle

Level 2 · Page 50

197

What is the phase relation in a twin cable acting like a coil?

A They are in phase
B The current leads the voltage
C The voltage leads the current

Level 2 · Page 54

198

Why does hardly any radiation occur from a twin cable?

A The H and E fields are in phase
B The H and E fields stay directly near the line
C The H and E fields cancel completely

Level 2 · Page 54

199

What results from spreading the lines of a twin cable?

A A loop antenna
B A dipole antenna
C A coaxial cable

Level 2 · Page 56

200

What is the field in the vicinity of a transmitting antenna known as?

A The radiation field
B The long distance field
C The short distance field

Level 2 · Page 58

201

What do the displacement currents produce according to Maxwell?

A A magnetic field
B A static electric charge
C A direct current

Level 2 · Page 58

202

What defines the polarization of a radio wave?

A The direction of the magnetic field
B The direction of the electrical field
C The direction of propagation

Level 2 · Page 60

203

How are aviation navigation and communication signals polarized?

A Both are vertically polarized
B Navigation vertical, communication horizontal
C Navigation horizontal, communication vertical

Level 2 · Page 60

204 **Which antennas transmit circular polarized waves?**

A Helical antennas
B Yagi antennas
C Loop antennas

Level 2 · Page 60

205 **What does circular polarization do?**

A It continuously alters its polarization
B It stays vertical at all times
C It stays horizontal at all times

Level 2 · Page 60

206 **Where does a λ/2 dipole emit?**

A Equally in all directions in space
B Along its axis, virtually zero at 90°
C At 90° to its axis, virtually zero along its axis

Level 2 · Page 64

207 **Which distance is the short distance field practically gone after?**

A 3 to 4 λ
B 30 to 40 λ
C 0.1 λ

Level 3 · Page 58

208 **What type of energy does the short distance field contain?**

A No energy
B Reactive energy
C Active energy

Level 3 · Page 58

209 **How are E and H related in the long distance field of an antenna?**

A 90° displaced, holding reactive energy
B In phase, transmitting active energy
C 180° out of phase, cancelling out

Level 3 · Page 58

210 **How does the sum of the Poynting vector change?**

A It falls in direct proportion to distance only
B It is constant in all directions
C It falls with distance squared and depends on direction

Level 3 · Page 62

NOT HIGHLIGHTED

211 **What does the radiation pattern of an antenna show?**

A The current along the antenna
B The impedance over frequency
C The distribution of radiation in different directions

Level 1 · Page 62

212 **Why can antenna theory be explained using a transmitting antenna?**

A Both antenna kinds are electrically treated the same way
B Receiving antennas have no fields
C Transmitting antennas are always smaller

Level 2 · Page 48

213 **Why do transmitting and receiving antennas differ mechanically?**

A Because of varying high transmission power
B Because of different frequencies
C Because of different polarizations

Level 2 · Page 48

214 **Which criteria decide the choice of antenna?**

A Weight, colour, cost and mounting
B Cable length, connector type and voltage
C Frequency range, bandwidth, propagation and polarization

Level 2 · Page 48

215 **Why is the antenna size often determined experimentally?**

A The formulas are unknown
B Antennas are always custom-made
C Surroundings can strongly influence it

Level 2 · Page 48

216 **What determines the strength and direction of the magnetic field around the twin cable?**

A The voltage
B The current
C The insulation

Level 2 · Page 52

217 **Where is the voltage highest on the twin cable?**

A Midway along the wires
B At the end of the wires
C At the feed point

Level 2 · Page 54

218 **Where is the electrical field strongest on the twin cable?**

A At the middle of the cable
B At the feed point
C At the end of the cable

Level 2 · Page 54

219 **What happens to the current and voltage distributions when the twin lines are spread?**

A They keep their sine shape
B They become rectangular
C They both become zero

Level 2 · Page 56

220 **Which distance range is the short distance field found within?**

A r < 0.1 λ
B r > 3–4 λ
C r = 1 λ

Level 2 · Page 58

221 **What is the phase relation of H and E in the short distance field?**

A 180° displaced
B 90° displaced
C In phase

Level 2 · Page 58

222 **What does the energy of the radiated electromagnetic field consist of?**

A The energy of the electrical field only
B The energy of the magnetic field only
C The energy of both fields

Level 2 · Page 58

223 **What gives the direction of propagation of the radiated energy?**

A The Poynting vector S
B The electrical field E
C The magnetic field H

Level 2 · Page 58

224 **What is the unit of the Poynting vector?**

A A/m
B V/m
C W/cm²

Level 2 · Page 59

225 **What polarization does a horizontally aligned dipole transmit?**

A Vertical
B Horizontal
C Circular

Level 2 · Page 60

226 **How are home radio and TV signals polarized?**

A Both horizontal
B Radio vertical, TV horizontal
C Radio horizontal, TV vertical

Level 2 · Page 60

227 **What is the theoretical reference antenna that radiates equally in all directions called?**

A A λ/2 dipole
B An isotropic radiator
C A Marconi antenna

Level 2 · Page 62

228 **What shape does the dipole radiation have from a side view?**

A A cardioid
B A circle
C A figure eight

Level 2 · Page 64

229 **What 3D shape is formed by the dipole radiation?**

A A donut
B A cone
C A sphere

Level 2 · Page 64

230 **How does a dipole emit when seen from the top?**

A Equally in all directions
B Only forward and back
C Only along its axis

Level 2 · Page 64

231 **How does a receiving dipole's characteristic compare with a transmitting dipole's?**

A It is the reverse
B It is the same
C It is circular

Level 2 · Page 64

232 **How is the Poynting vector oriented to the E and H fields?**

A Parallel to H
B Parallel to E
C At right angles to both

Level 3 · Page 58

Coverage All text on pages 48–65 covered. Page 49 figure is covered by the Yagi questions in T8. Pages 51, 53, 55, 57, 63 and 65 are figures.
Conflict Page 52: the text reads 'the typical length is /4' with the λ missing from the text layer; the figure on page 53 shows λ/4.

8 · Antenna Types

Pages 66-82 · 52 questions

● **HIGHLIGHTED**

● **233** **What does a Yagi antenna consist of?**

A A dipole plus one or more directors
B A loop plus one or more reflectors
C Two dipoles connected in parallel

Level 1 · Page 67

● **234** **What are the directors of a Yagi antenna often referred to as?**

A Feed elements
B Parasitic elements
C Active elements

Level 1 · Page 67

● **235** **What is the rod antenna also referred to as?**

A Yagi antenna
B Slot antenna
C Marconi antenna

Level 1 · Page 69

● **236** **What is the loop antenna used for?**

A Transmitting high frequencies
B Receiving long and medium wave
C Receiving UHF signals

Level 1 · Page 71

● **237** **Which are examples of surface radiators?**

A The Yagi antenna and the dipole
B The horn antenna and the lens antenna
C The rod antenna and the loop antenna

Level 1 · Page 73

● **238** **What is a horn antenna in effect?**

A A funnel-shaped waveguide
B A short rod with a ground plane
C A coiled wire loop

Level 1 · Page 73

● **239** **What is the impedance of free air?**

A 150 Ω, the twin cable value
B 50 Ω, the coax standard
C 120π, about 377 Ω

Level 1 · Page 73

240 **Which frequencies are horn antennas mainly used for?**

A Below 30 MHz
B 1 GHz and above
C Between 3 MHz and 30 MHz

Level 1 · Page 73

241 **Where is the radiating element of a reflector antenna installed?**

A At the rim of the parabolic reflector
B At the focal point of a parabolic reflector
C Behind the parabolic reflector

Level 1 · Page 76

242 **What else is a delay lens antenna known as?**

A A dielectric lens
B A parabolic lens
C A metal plate lens

Level 1 · Page 79

243 **Which dipole types are compared in the slot antenna figure?**

A Yagi and Marconi dipole
B Electrical dipole and magnetic dipole
C Folded dipole and loop

Level 1 · Page 82

244 **How symmetric is the field of a dipole antenna?**

A It is perfectly symmetric
B It is not symmetric at all
C It is symmetric only at night

Level 2 · Page 69

245 **Why is the loop antenna especially suitable as a Direction-Finding antenna?**

A Because of its directional characteristic
B Because of its very high gain
C Because of its circular polarization

Level 2 · Page 71

246 **Which use is the loop antenna NOT suitable for?**

A As a receiving antenna
B As a direction-finding antenna
C As a transmitting antenna

Level 2 · Page 71

247 **What does the funnel shape of a horn antenna act as?**

A A filter for unwanted frequencies
B An amplifier for the waveguide signal
C A transformer between waveguide and free-air impedance

Level 2 · Page 73

248 **What is needed to achieve high antenna gain and a high degree of bunching?**

A Large apertures, small flare angles, long lengths
B Short lengths, small apertures, wide angles
C Small apertures, wide flare angles, short lengths

Level 2 · Page 76

249 **How can low bunching in short horn antennas be overcome?**

A By using lens and reflector antennas
B By using a longer coaxial cable
C By using circular polarization

Level 2 · Page 76

250 **What does the focusing angle θ of a reflector antenna depend on?**

A The power and the feed line length
B The polarization and the reflector colour
C The frequency and the reflector diameter

Level 2 · Page 76

251 **What is the slot antenna equivalent to?**

A A λ/4 rod on a ground plane
B The magnetic equivalent of the electrical dipole
C The electrical equivalent of the loop

Level 2 · Page 81

252 **What gives a Yagi antenna its increased gain in the antenna direction?**

A The greater length of the dipole
B Superposition of the dipole and parasitic element fields
C The circular polarization of the dipole

Level 3 · Page 67

253 **How does the length of a parasitic element decide its function?**

A Any length acts as a director
B Shorter than the dipole: reflector; longer: director
C Shorter than the dipole: director; longer: reflector

Level 3 · Page 67

254 **How does the field of a λ/4 rod antenna relate to the field of a λ/2 dipole?**

A It corresponds to one half of the dipole field
B It is twice the dipole field
C It is identical to the dipole field

Level 3 · Page 69

255 **What does the induced voltage U0 of a loop antenna depend on?**

A Wire diameter, frequency and polarization
B Loop shape, cable length and power
C Loop area, number of windings and angle θ

Level 3 · Page 71

256 **What increases the gain of a horn antenna?**

A A wider flare angle above 40°
B Faster widening and a shorter funnel
C More gradual widening and a longer funnel

Level 3 · Page 73

257 **What does a convex dielectric lens with εr > 1 achieve?**

A Centre beams radiate before edge beams
B Centre and edge beams radiate in phase
C Edge beams are absorbed by the lens

Level 3 · Page 76

258 **How do the beams leave a delay lens with the correct delay?**

A At the same time and in parallel
B At different times and diverging
C At the same time and converging

Level 3 · Page 76

NOT HIGHLIGHTED

259 **Which antenna types are shown as examples on the antenna types page?**

A Horn, slot, Yagi-Uda and patch
B Dish, helix, whip and wire
C Loop, rod, umbrella and T

Level 1 · Page 66

260 **What does a loop antenna consist of?**

A A dipole with parasitic elements
B One or more square or round wire windings
C A funnel-shaped waveguide

Level 1 · Page 71

261 **What is the slot antenna also referred to as?**

A An electrical dipole
B A magnetic dipole
C A Marconi antenna

Level 1 · Page 81

262 **How is a slot antenna formed?**

A By a short rod on a ground plane
B By a coil of wire with several turns
C By a slot in a metal sheet or waveguide

Level 1 · Page 81

263 **What happens to the parasitic elements when the Yagi dipole transmits?**

A They absorb all the energy
B They are excited and transmit too
C They block the dipole's field

Level 2 · Page 67

264 **What creates the phase shift between the dipole and the parasitic elements?**

A The polarization of the dipole
B The distance to the dipole
C The length of the feed line

Level 2 · Page 67

265 **What happens to the Yagi antenna gain on the sides?**

A It is unchanged
B It is increased
C It is reduced

Level 2 · Page 67

266 **How does a Yagi antenna behave when used for reception?**

A The effects are reversed
B It has no gain
C The same effects apply

Level 2 · Page 67

267 **What element spacing is shown in the Yagi figure?**

A 0.15 λ
B 0.5 λ
C 1 λ

Level 2 · Page 68

268 **What happens if the plane has finite conductivity?**

A The field changes accordingly
B The field stays fully symmetric
C The field doubles in strength

Level 2 · Page 69

269 **Where is the rod antenna used?**

A Where very high gain is needed
B Where circular polarization is needed
C Where the antenna must be as small as possible

Level 2 · Page 69

270 **Which component of the field induces the voltage in a loop antenna?**

A The electrical component
B The magnetic component
C The Poynting vector

Level 2 · Page 71

271 **How do surface radiators differ from wire antennas?**

A A surface is also involved in radiation
B They do not radiate at all
C They use only a thin wire

Level 2 · Page 73

272 **What do the flare angles of a horn antenna depend on?**

A Frequency and power only
B Polarization and gain
C Height, width and length of the horn

Level 2 · Page 74

273 **What does the propagation velocity of a wave in a lens depend on?**

A The permittivity εr
B The polarization
C The reflector diameter

Level 2 · Page 76

274 **How are the reflected waves of a reflector antenna in the aperture plane?**

A 90° out of phase
B 180° out of phase
C In phase

Level 2 · Page 76

275 **How many beams does a dish antenna produce compared with a lens antenna?**

A Dish: multiple beams; lens: single beam
B Dish: single beam; lens: multiple beams
C Both produce multiple beams

Level 2 · Page 78

276 **Which principle do lens antennas use?**

A The diverging lens, to spread the energy
B The converging lens, to collimate energy into plane waves
C The parabolic mirror, to reflect the energy

Level 2 · Page 79

277 **What length is a Yagi director compared with the λ/2 dipole?**

A λ/4
B λ/2 + 5 %
C λ/2 − 5 %

Level 3 · Page 68

278 **What length is a Yagi reflector compared with the λ/2 dipole?**

A λ/2 + 5 %
B λ
C λ/2 − 5 %

Level 3 · Page 68

279 **What happens to the field above a perfectly conductive plane placed in the middle of a dipole?**

A It stays the same, even without the lower half
B It doubles in strength above the plane
C It disappears completely above the plane

Level 3 · Page 69

280 **When is the maximum voltage induced in a loop antenna?**

A Loop plane at 90° to the propagation
B Loop plane parallel to the ground
C Loop plane in line with the propagation

Level 3 · Page 71

281 **Which horn dimensions give favourable focusing?**

A Flare angle of 90°, length equal to λ
B Flare angle under 40°, length and width large to λ
C Flare angle over 40°, length and width small to λ

Level 3 · Page 73

282 **Which beams are delayed most by a delay lens?**

A Beams that hit the lens centre
B Beams that hit the lens edge
C All beams equally

Level 3 · Page 76

283 **Why does a wave undergo retardation in a delay lens?**

A The lens reduces the frequency
B The lens medium increases the electrical path length
C The lens reflects part of the wave

Level 3 · Page 79

284 **What polarization does a vertically installed slot antenna produce?**

A Circular
B Horizontal
C Vertical

Level 3 · Page 81

Coverage All text on pages 66–82 covered. Page 75 and 80 are pictures (lens simulations) with no testable text.
Conflict Page 78 says the lens antenna 'focuses RF waves thru Reflection'. A lens works by refraction (page 79 also describes it as a converging lens). No question was written on that line.

9 · Antenna Characteristics and Antenna Tuning

Pages 83-89 · 29 questions

● **HIGHLIGHTED**

● **285** **What does the lobe width indicate?**

A The bandwidth of an antenna
B The impedance of an antenna
C The directivity of an antenna

Level 1 · Page 83

● **286** **What is the usual reference antenna for antenna gain?**

A An isotropic antenna
B A Yagi antenna
C A λ/4 rod antenna

Level 1 · Page 83

● **287** **What unit is antenna gain normally given in?**

A Ohms
B Decibels
C Watts

Level 1 · Page 83

● **288** **What is the effect of concentrating the radiation in one direction?**

A Lower field strength there for the same power
B No change in field strength
C Greater field strength there for the same power

Level 2 · Page 83

● **289** **What is antenna gain G?**

A Power lost along the antenna feed line
B Ratio of the antenna length to the wavelength
C Radiated power gain in a direction over a reference

Level 2 · Page 83

● **290** **How is antenna gain gauged?**

A In comparison to the receiver noise
B In comparison to a reference source
C In comparison to the feed line loss

Level 2 · Page 85

● **291** **How is an antenna tuned mechanically?**

A Its length is adjusted to the wavelength
B Its feed line is shortened
C Its polarization is changed

Level 2 · Page 86

292 **How can an antenna of limited length still be brought to resonance?**

A By using a loop instead
B With an L or T shaped tip or an umbrella
C By lengthening the feed line

Level 2 · Page 86

293 **How must the length of an antenna be adjusted when it is tuned?**

A Only mechanically to the power
B Only electrically to the polarization
C Mechanically or electrically to the frequency

Level 2 · Page 88

294 **What diagram shows the ohmic resistance and reactance of an antenna over frequency?**

A The locus
B The radiation pattern
C The resonance curve

Level 2 · Page 88

295 **What must the reactance be to tune the antenna to the desired frequency?**

A Zero
B Equal to R
C Maximum

Level 2 · Page 88

296 **What tunes the antenna over the frequency range shown in the locus figure?**

A The power amplifier
B The HF coupler
C The radio altimeter

Level 2 · Page 89

297 **What is the effect of the end capacitance on the antenna?**

A Higher resonant frequency, shorter electrically
B Lower resonant frequency, longer electrically
C Same resonant frequency, same length

Level 3 · Page 86

NOT HIGHLIGHTED

298 **What does the locus show on its X-axis and Y-axis?**

A X: ohmic resistance; Y: reactance
B X: frequency; Y: gain
C X: reactance; Y: ohmic resistance

Level 1 · Page 88

299 **What angle does the lobe width give?**

A Apex angle where power falls to half
B Angle of maximum radiated power
C Angle where the radiation is zero

Level 2 · Page 83

300 **What is the minimum received power within the lobe width of a receiving antenna?**

A 0.25 PE max
B 0.707 PE max
C 0.5 PE max

Level 2 · Page 83

301 **Which dB point is the lobe width equivalent to?**

A The 3 dB point
B The 10 dB point
C The 6 dB point

Level 2 · Page 83

302 **What is the effect of using a directional antenna?**

A The same as reducing the frequency
B The same as increasing transmitter power
C The same as using a longer feed line

Level 2 · Page 83

303 **Which antennas are often used as reference antennas in practice?**

A Loop antennas
B λ/4 rods
C λ/2 dipoles

Level 2 · Page 83

304 **What is the antenna boresight?**

A The axis of maximum gain
B The axis of zero radiation
C The axis of the feed line

Level 2 · Page 85

305 **What beamwidth typically gives decent coverage at one metre in compliance testing?**

A 90 degrees
B 3 degrees
C 30 degrees

Level 2 · Page 85

306 **What is the gain of an isotropic antenna?**

A 3 dBi
B 1.64 dBi
C 0 dBi

Level 2 · Page 85

307 **What is the gain of a λ/2 dipole compared to an isotropic antenna?**

A 1.64
B 3.0
C 1.0

Level 3 · Page 83

308 **Why must a long wave antenna be shorter than λ/4?**

A Long wave antennas need high gain
B Long wave wavelengths can be several kilometres
C Long wave needs circular polarization

Level 3 · Page 86

309 **How is the end capacitance CE connected relative to the antenna capacitance CA?**

A In series, giving a smaller total capacitance
B It replaces CA completely
C In parallel, giving a larger total capacitance

Level 3 · Page 86

310 **How is an antenna that is too short for f1 tuned?**

A With a capacitor in series
B With an inductor in parallel
C With an inductor in series

Level 3 · Page 88

311 **How is an antenna that is too long for f2 tuned?**

A With a capacitor in parallel
B With a capacitor in series
C With an inductor in series

Level 3 · Page 88

312 **How is an antenna that is too short for f3 tuned?**

A With a capacitor in parallel
B With an inductor in parallel
C With a capacitor in series

Level 3 · Page 88

313 **How is an antenna that is too long for f4 tuned?**

A With a capacitor in parallel
B With an inductor in series
C With an inductor in parallel

Level 3 · Page 88

Coverage All text on pages 83–89 covered. Page 84 figure (lobe width, −3 dB) covered by the lobe width questions. Page 87 is a figure.

10 · RF Lines, Matching and Standing Waves

Pages 90-103 · 45 questions

● **HIGHLIGHTED**

● **314** **What are radio frequency lines required for?**

A Transferring RF energy with as little loss as possible
B Generating the basic radio frequency
C Converting RF energy into free radiation

Level 1 · Page 90

● **315** **Which are other applications of RF lines?**

A Polarization, focusing and filtering
B Resonant circuits, resistance transformation and matching
C Modulation, detection and amplification

Level 1 · Page 90

● **316** **What type of line is a coaxial cable?**

A An unshielded line
B An asymmetric line
C A symmetric line

Level 1 · Page 90

● **317** **What impedance do typical symmetric twin cables start at?**

A About 150 Ω
B About 50 Ω
C About 377 Ω

Level 1 · Page 94

● **318** **What is the matching principle known as in electrical engineering?**

A Thomson's oscillation formula
B Kirchhoff's first law
C Power transfer theorem or Jacobi's Law

Level 1 · Page 96

● **319** **How are RF lines differentiated electrically and physically?**

A Electrically symmetric/asymmetric; physically shielded/unshielded
B Electrically matched/mismatched; physically open/shorted
C Electrically shielded/unshielded; physically symmetric/asymmetric

Level 2 · Page 90

● **320** **What are the features of a symmetric cable?**

A A shield around a concentric inner line
B One line grounded, one line at RF potential
C Equal capacity to ground, equal and opposite RF signals

Level 2 · Page 90

321 **What are the features of an asymmetric cable?**

A Both lines with equal capacity to ground
B One line at RF potential, the other grounded
C Both lines at equal RF potential

Level 2 · Page 90

322 **What determines the electrical data of RF lines?**

A The physical construction and the dielectric
B The applied frequency and power
C The antenna type and polarization

Level 2 · Page 92

323 **What determines the ratio of ix and ux on a line?**

A The series and parallel resistance
B The length of the line
C The line inductance L' and capacity C'

Level 2 · Page 92

324 **How is the impedance of a cable generally calculated?**

A Z0 = √(C/L)
B Z0 = √(L/C)
C Z0 = L × C

Level 2 · Page 94

325 **What are the disadvantages of coax cables with Teflon or polyethylene dielectric?**

A Lower dampening, higher voltage rating, higher power
B High price and difficult handling
C Higher dampening, lower voltage rating, lower power

Level 2 · Page 94

326 **What is required to transfer as much energy as possible from transmitter to antenna?**

A The transmitter and antenna impedances must match
B The line must be as long as possible
C The antenna must be longer than λ

Level 2 · Page 96

327 **What is created when an antenna with a non-matching impedance is connected to an RF line?**

A A sky wave
B A standing wave
C A matched line

Level 2 · Page 98

328 **What does the standing wave ratio indicate?**

A The physical length of the RF line
B The antenna-to-line impedance difference
C The power radiated by the antenna

Level 2 · Page 98

329 **What does an SWR of 1 mean?**

A Identical impedances, maximum power transfer
B Open line, no power transfer
C Shorted line, maximum reflection

Level 2 · Page 98

330 **What is the SWR of an open or shorted line?**

A 0
B 1
C Infinite

Level 2 · Page 98

331 **Which two kinds of reflection are shown in the standing wave figure?**

A Ground and sky reflection
B Opposite phase and in-phase reflection
C Hard and soft absorption

Level 2 · Page 99

332 **What appears on a shorted or open-ended line?**

A No current and no voltage
B Large standing waves for current and voltage
C A perfectly matched load

Level 2 · Page 102

333 **Why is one line capacity (C20) to ground missing in an asymmetric cable?**

A Ground itself is the conductor
B The line is too short
C The line is shielded

Level 3 · Page 90

334 **Which capacities does the inner wire of a coaxial cable have?**

A C10 to ground, no C12 to the shield
B Equal C10 and C20 to ground
C C12 against the shield, no C10 to ground

Level 3 · Page 90

335 **What creates a hard reflection on an RF line?**

A A short or an open line
B A small impedance deviation
C A perfectly matched load

Level 3 · Page 98

336 **What happens at the end of an open RF line?**

A Maximum current flows, the voltage is zero
B Current and voltage are both zero
C No current flows, the voltage is at maximum

Level 3 · Page 100

337 **What happens at the end of a shorted RF line?**

A No current flows, the voltage is at maximum
B Current and voltage are both at maximum
C Maximum current flows, the voltage is zero

Level 3 · Page 100

338 **How does a shorted λ/4 line behave for the supply generator?**

A As a series resonant circuit
B As a parallel resonant circuit
C As a pure inductor

Level 3 · Page 102

339 **How does an open-ended λ/4 line behave for the supply generator?**

A As a series resonant circuit
B As a parallel resonant circuit
C As a pure capacitor

Level 3 · Page 102

NOT HIGHLIGHTED

340 **What impedances are common for coaxial cables?**

A 377 Ω
B 150 Ω to 300 Ω
C 50 Ω to 75 Ω

Level 1 · Page 94

341 **What is the standard impedance of coaxial power cables in modern installations?**

A 50 Ω
B 75 Ω
C 150 Ω

Level 1 · Page 94

342 **What are the advantages of coax cables with Teflon or polyethylene dielectric?**

A Low dampening and high power
B Easy to handle and reasonably priced
C High voltage rating

Level 1 · Page 94

343 **Which values does every line have per unit length?**

A Only the line capacity C'
B L', C', series and parallel resistance
C Only the line inductance L'

Level 2 · Page 92

344 **Which line values can be ignored in most practical cases?**

A Series and parallel resistance
B Line inductance and capacity
C Line capacity C' alone

Level 2 · Page 92

345 **What determines the line impedance Z0?**

A The power applied to the line
B U/I ratio of an infinitely long line
C The physical length of the line

Level 2 · Page 94

346 **What does Z0 depend on?**

A The applied frequency only
B The cable's materials and dimensions
C The transmitter power only

Level 2 · Page 94

347 **What must the HF lines between transmitter and antenna have?**

A A matching impedance throughout
B The lowest possible impedance
C The highest possible impedance

Level 2 · Page 96

348 **What load resistance gives a matched RF line?**

A RL = 0
B RL = 2 Z0
C RL = Z0

Level 2 · Page 97

349 **Which materials reflect waves well in the pond example?**

A Soft materials like clay
B Soft materials like sand
C Hard materials like concrete

Level 2 · Page 98

350 **What happens when the reflected wave meets the initial wave?**

A They superpose, amplify or cancel
B They both stop at that point
C They pass with no effect at all

Level 2 · Page 98

351 **What does a greater SWR indicate?**

A A better match of impedances
B Smaller impedance deviation
C Greater impedance deviation

Level 2 · Page 98

352 **How is the SWR calculated from voltage?**

A E minimum ÷ E maximum
B E maximum ÷ E minimum
C E maximum × E minimum

Level 2 · Page 98

353 **How is the SWR worked out from the load and line impedances?**

A Zload − Z0, whichever is greater
B Zload × Z0, whichever is greater
C Zload ÷ Z0 or Z0 ÷ Zload, whichever is greater

Level 2 · Page 98

354 **What damage can a very high voltage on an RF line cause?**

A Arcing and sparking in the dielectric
B Loss of polarization
C Burned spots from heating

Level 2 · Page 100

355 **What damage can a very high current on an RF line cause?**

A A change of impedance only
B Heated spots, burns or small fires
C Arcing in the dielectric

Level 2 · Page 100

356 **Which spacing does in-phase superposition occur at on a hard-reflecting line?**

A Every full wavelength
B Every quarter wavelength
C Every half wavelength

Level 3 · Page 100

357 **How does an open λ/2 line behave for the supply generator?**

A As a parallel resonant circuit
B As a series resonant circuit
C As a matched load

Level 3 · Page 102

358 **Where are the current node and voltage loop at the input of a line acting as a parallel resonant circuit?**

A Current node and voltage loop at the input
B Both nodes at the input
C Current loop and voltage node at the input

Level 3 · Page 102

Coverage All text on pages 90–103 covered. Pages 91, 93, 95, 97, 99, 101, 103 are figures; labels used where testable.

11 · Resonant Circuits

Pages 105-117 · 42 questions

● **HIGHLIGHTED**

● **359** **What does a mechanical oscillator consist of?**

A A spring and a resistor only
B A mass and a return force
C A coil and a capacitor

Level 1 · Page 105

● **360** **What do electric oscillator systems consist of?**

A Resistors and diodes
B Transformers and resistors
C Inductors and capacitors

Level 1 · Page 105

● **361** **What is the bandwidth B of a resonant circuit?**

A The frequency range where the curve drops to a set value each side of the peak
B The total range of frequencies the circuit can pass
C The frequency at the peak of the resonance curve

Level 1 · Page 114

● **362** **What are resonant circuits used for?**

A Generation of a steady DC voltage
B Selective amplification, muting, transformation
C Rectification of AC into DC voltage

Level 1 · Page 116

● **363** **How is energy stored in an LC oscillator?**

A Inductor as voltage, capacitor as a magnetic field
B Both as heat in the resistor
C Inductor as a magnetic field, capacitor as voltage

Level 2 · Page 105

● **364** **What happens in a series resonant circuit at the resonant frequency f0?**

A XL = XC, iL and iC cancel, only iR flows
B XL > XC and the circuit is inductive
C XL = XC, UL and UC cancel, high current limited by RS

Level 2 · Page 109

● **365** **What determines the bandwidth?**

A The quality factor Q
B The applied voltage
C The resistor value only

Level 2 · Page 114

366 **What is selectivity?**

A Passing the widest band of frequencies
B Suppressing frequencies other than the resonant one
C Raising the resonant frequency

Level 2 · Page 114

367 **How large can the voltages across L and C be in a series circuit at f0?**

A Always equal to the applied voltage
B Much higher than the applied voltage
C Always lower than the applied voltage

Level 3 · Page 109

368 **What happens in a parallel resonant circuit at the resonant frequency f0?**

A UL and UC cancel, a high current flows
B iL is greater and the circuit is inductive
C iL and iC cancel, only iR flows

Level 3 · Page 111

369 **How is a strong unwanted local transmitter suppressed?**

A Series circuit tuned to the wanted signal in the line
B Series circuit tuned to it as ground leg of a divider
C Parallel circuit tuned to it as ground leg of a divider

Level 3 · Page 116

370 **How is one desired frequency boosted at the receiver input?**

A Series circuit tuned to it as ground leg of a divider
B Parallel circuit tuned to the unwanted signal
C Parallel circuit tuned to it as ground leg of a divider

Level 3 · Page 116

NOT HIGHLIGHTED

371 **What are oscillating and resonant systems used for?**

A Storing DC energy for long periods
B Generating, amplifying or damping periodic processes
C Converting AC voltage into DC voltage

Level 1 · Page 105

372 **How does the impedance of a resistor change with frequency?**

A It stays constant
B It falls non-linearly
C It rises in proportion

Level 1 · Page 113

373 **Why does energy travel back and forth between inductor and capacitor?**

A They delay voltage against current
B They both behave as pure resistors
C They are fed from a DC source

Level 2 · Page 105

374 **What determines the frequency of a mechanical pendulum?**

A The amplitude only
B The external force
C The mass and the length l

Level 2 · Page 107

375 **What happens to the sum of potential and kinetic energy in a pendulum without losses?**

A It increases
B It decreases
C It stays constant

Level 2 · Page 107

376 **How is energy introduced into an LC oscillator?**

A A magnet is moved near the coil
B An external voltage charges the capacitor
C A resistor is placed in series

Level 2 · Page 107

377 **What determines the frequency of an LC oscillator?**

A The switch position
B The dimensions of L and C
C The applied voltage

Level 2 · Page 107

378 **What is Thomson's oscillation formula?**

A f0 = 1 / (2π√(LC))
B f0 = √(L/C) / 2π
C f0 = 2π√(LC)

Level 2 · Page 107

379 **Which mechanical energy matches the magnetic field energy in the oscillator figure?**

A Heat energy
B Potential energy
C Kinetic energy

Level 2 · Page 108

380 **What represents the losses of L and C in a series resonant circuit?**

A RS
B RP
C XL

Level 2 · Page 109

381 **What does the geometric sum of UL, UC and UR equal in a series circuit?**

A The applied voltage U
B The current i
C Zero at all times

Level 2 · Page 109

382 **When is the impedance Z of a series resonant circuit lowest?**

A Below f0
B Above f0
C At f0

Level 2 · Page 109

383 **What is the phase angle between U and i in a series circuit at resonance?**

A +90 degrees
B Zero degrees
C −90 degrees

Level 2 · Page 109

384 **What represents the losses of L and C in a parallel resonant circuit?**

A RS
B XC
C RP

Level 2 · Page 111

385 **What does the total current equal in a parallel circuit?**

A The geometric sum of the branch currents
B The current in the capacitor only
C Zero at all frequencies

Level 2 · Page 111

386 **What does the impedance Z of a parallel circuit equal at resonance?**

A Zero
B RP
C XL + XC

Level 2 · Page 111

387 **How does the impedance of an inductor change with frequency?**

A It stays constant
B It rises in proportion
C It falls non-linearly

Level 2 · Page 113

388 **How does the impedance of a capacitor change as frequency increases?**

A It stays constant
B It increases in proportion
C It decreases, not linearly

Level 2 · Page 113

389 **What is the frequency at which XL = XC called?**

A The resonant frequency fR
B The cut-off frequency
C The critical frequency

Level 2 · Page 113

390 **Which level is the normal bandwidth measured at?**

A −6 dB (0.5)
B −10 dB (0.1)
C −3 dB (0.707)

Level 2 · Page 114

391 **What must be done if a bandwidth other than −3 dB is used?**

A Nothing, it is standard
B It must be specified
C It must be doubled

Level 2 · Page 114

392 **How is the bandwidth worked out from the quality factor Q?**

A B = Q / f0
B B = f0 / Q
C B = f0 × Q

Level 2 · Page 114

393 **What happens to other frequencies when a series circuit suppresses one signal?**

A They are not affected
B They are boosted
C They are also shorted to ground

Level 2 · Page 116

394 **What happens at the desired frequency with a parallel circuit used for boosting?**

A The signal is shorted to ground
B Impedance and output voltage are high
C Impedance and output are both zero

Level 2 · Page 116

395 **When does the magnetic field in the LC oscillator reach maximum strength?**

A When the voltage is zero
B When the voltage is maximum
C When the switch is open

Level 3 · Page 107

396 **What charges the capacitor with reversed polarity in an LC oscillator?**

A Voltage induced by the coil's field
B The external voltage source
C The losses in the resistor

Level 3 · Page 107

397 **How does a series resonant circuit behave below and above f0?**

A Resistive at all frequencies
B Below inductive, above capacitive
C Below capacitive, above inductive

Level 3 · Page 109

398 **How does a parallel resonant circuit behave below and above f0?**

A Below inductive, above capacitive
B Below capacitive, above inductive
C Resistive at all frequencies

Level 3 · Page 111

399 **What is the trade-off of a high selectivity?**

A It has no trade-off
B It comes at the cost of bandwidth
C It comes at the cost of gain

Level 3 · Page 114

400 **What does a large Q factor do to the resonance curve?**

A Makes it gradual
B Makes it flat
C Makes it sharp

Level 3 · Page 115

Coverage All text on pages 105–117 covered. Pages 106, 108, 110, 112, 115 and 117 are figures; page 108 and 115 labels used.
Conflict Page 105: 'displaces' is a typo for 'displaced'. Page 107 spells the formula name 'Thompsons'; questions use 'Thomson'.

Answer Key

All 400 questions · Key balance A 134 · B 133 · C 133

1 · Radio Technology Basics

#	Ans	Why (LTT wording)	Page
1	**B**	Both transmitting and receiving equipment consist basically of electronic power supplies, amplifiers and oscillators.	6
2	**C**	A basic radio communication system may consist of only a transmitter and a receiver.	6
3	**A**	The stage required to place the audio intelligence on the RF signal is the modulator.	6
4	**C**	Sonic waves are converted to electrical signals by microphones.	6
5	**A**	Loudspeakers convert electrical signals to sonic waves.	6
6	**C**	They are connected by the medium through which the electromagnetic waves travel.	6
7	**A**	No medium is necessary for the propagation of electromagnetic waves.	6
8	**A**	The electromagnetic variations are propagated through the medium (space) from the transmitting antenna to the receiving antenna.	6
9	**B**	Radio equipment can be divided into transmitting equipment and receiving equipment.	6
10	**A**	The transmitter comprises an oscillator, RF amplifiers and the modulator stage.	6
11	**B**	The oscillator generates a basic radio frequency.	6
12	**C**	Media named: wire, wave guide, coaxial cable, nonconducting fluids or gases, air or vacuum.	6
13	**A**	Figure 1: microphone → transmitter → antennas → receiver → loudspeaker.	7
14	**C**	The receiving antenna converts the received energy into a flow of alternating radio frequency currents.	6
15	**B**	The receiver converts these current variations into the intelligence contained in the transmission.	6

2 · AC Properties, Frequency and Angular Frequency

#	Ans	Why (LTT wording)	Page
16	**B**	In communications engineering both AC as well as DC is used.	8
17	**A**	The time for one cycle to complete is called cycle time T.	8
18	**C**	The symbol for period is T.	8
19	**C**	The period is measured in seconds (s).	8
20	**A**	The frequency of AC is the number of complete cycles, i.e. oscillations, in one second.	10
21	**B**	1 cps = Hz.	10
22	**A**	$\omega = 2\pi \times f$.	12
23	**B**	The current I varies in direction (polarity) and in quantity over a period of time t.	8

#	Ans	Why (LTT wording)	Page
24	**B**	The polarity and current magnitude change then repeats periodically. The same applies to the voltage.	8
25	**C**	Figure 2: 1 cycle = positive alteration + negative alteration. Page 8: the positive and negative half-wave together constitute one AC cycle.	9
26	**C**	Frequency is also defined as the reciprocal of the cycle time T: f = 1/T.	10
27	**B**	The rotational speed of the tip depends on the RPM and the diameter of the generator.	12
28	**A**	Its length is given by the amplitude of the voltage while the RPM is determined by the frequency.	12
29	**B**	Angular velocity is a vector quantity, while angular frequency is a scalar quantity.	13
30	**B**	AC is the type of current used in the mains supply in the field of power engineering.	8
31	**A**	The abbreviation for alternating current is AC.	8
32	**C**	Every generator containing rotating parts will create a sinusoidal AC.	8
33	**C**	The sinusoidal waveform is the standard for AC.	8
34	**B**	The symbol for frequency is f.	10
35	**C**	The frequency is measured in hertz (Hz).	10
36	**A**	Very often frequency is expressed in kHz, MHz or GHz.	10
37	**C**	A sinus typically is created by a rotational movement, such as a wire loop rotating in a magnetic field or a magnet rotating in a generator.	12
38	**B**	Both are represented by ω.	13
39	**A**	In the positive half-wave the voltage goes from the 0-reference line up to the positive peak and from +peak down to 0.	8
40	**B**	The voltage rises from 0 to a negative peak value and falls from −peak to the 0-reference line.	8
41	**C**	The higher the frequency the shorter the cycle time; the lower the frequency the longer the cycle time.	10
42	**A**	[f] = 1/s = s^{-1}.	10
43	**C**	With a length of 1, the rotational speed only depends on the RPM.	12
44	**A**	The formula for both angular velocity and angular frequency is the same.	13

3 · Harmonic, Triangular and Sawtooth Waves

#	Ans	Why (LTT wording)	Page
45	**B**	Multiples of the basic frequency are further harmonic waves.	16
46	**C**	The sinusoidal wave with the same frequency as the non-sinusoidal wave is called basic oscillation or 1st harmonic.	16
47	**B**	By superimposing the 1st, 3rd and 5th harmonic, an almost rectangular waveform is obtained.	16
48	**A**	For triangular waves the odd harmonics have to be used, just like with the rectangular ones.	19
49	**B**	For sawtooth waves all harmonics are required, the odd ones and the even ones.	19

#	Ans	Why (LTT wording)	Page
50	**A**	Besides the rectangular type, triangular and sawtooth waves are also possible.	19
51	**A**	Non-sinusoidal oscillations can be resolved to sinus-shape by mathematics or graphic means.	16
52	**C**	Rectangular waves can be resolved to an infinite number of uneven harmonics.	16
53	**B**	Harmonic numbers and amplitudes are determined by the shape of the non-sinusoidal oscillation.	16
54	**B**	The red line indicates a square-wave that would result from summating an infinite number of odd harmonics.	18
55	**C**	The amplitude of a harmonic is the amplitude of the 1st harmonic divided by the number of the actual harmonic.	16
56	**C**	For triangular waves the amplitude decreases with the square of the number of the harmonic.	19
57	**A**	For sawtooth waves the amplitude decreases with the number of the harmonic.	19

4 · Frequency Bands, Wavelength and Aviation Frequencies

#	Ans	Why (LTT wording)	Page
58	**B**	The three most important frequency bands are HF, VHF and UHF.	21
59	**A**	HF high frequency: 3 MHz – 30 MHz.	21
60	**C**	VHF very high frequency: 30 MHz – 300 MHz.	21
61	**B**	UHF ultra-high frequency: 300 MHz – 3000 MHz.	21
62	**A**	Figure 7: Communication – Data Communication, Radio Broadcast, Satellite Broadcast.	22
63	**C**	In free space (vacuum) it would propagate with the speed of light c.	23
64	**A**	$\lambda = c \times T = c / f$ [m].	23
65	**A**	In a vacuum the speed of light c would be 299 792 km/s or approximately 300 000 km/s.	23
66	**C**	Radio frequencies up to 300 GHz are divided into decadic ranges.	27
67	**A**	Each decade is designated as a frequency band.	27
68	**B**	The frequencies used for radar technology are divided into special radar bands.	27
69	**B**	Figure 9: VHF Com 118–137 MHz.	28
70	**C**	Figure 9: HF Com 2–30 MHz.	28
71	**B**	Figure 9: UHF Com 225–400 MHz.	28
72	**A**	Figure 9: ADF 200–1750 kHz.	28
73	**C**	Figure 9: SatCom 1,5 GHz.	28
74	**B**	Figure 9: Weather Radar 9,375 GHz.	28
75	**A**	Figure 9: VLF Navigation 10–24 kHz.	28
76	**A**	Figure 9: Identification Signals – DME: 1350 Hz.	28
77	**B**	Figure 9: Identification Signals – VOR, ILS, ADF: 1020 Hz.	28
78	**C**	Figure 9: Voice 300 Hz – 3 kHz.	28

#	Ans	Why (LTT wording)	Page
79	**C**	Figure 9: Marker: 400, 1300, 3000 Hz.	28
80	**B**	The propagation speed depends on the medium used.	23
81	**C**	The higher the density of the medium the lower the propagation speed will be.	23
82	**B**	RF = Radio Frequency (carrier wave).	27
83	**A**	IF = Intermediate Frequency.	27
84	**C**	AF = Audio Frequency.	27
85	**B**	Radar band table: X 5200–10 900 MHz.	27
86	**C**	Radar band table: L 390–1550 MHz.	27
87	**A**	Radar band table: S 1550–5200 MHz.	27
88	**A**	Radar band table: C 3900–6200 MHz.	27
89	**C**	Radar band table: K 10 900–36 000 MHz.	27
90	**C**	Figure 9: 300 MHz ↔ 1 m.	28
91	**A**	Figure 9: LOC 108–112 MHz.	28
92	**A**	Figure 9: VOR 112–118 MHz.	28
93	**B**	Figure 9: GS 329,3–399,95 MHz.	28
94	**A**	Figure 9: Marker 75 MHz.	28
95	**B**	Figure 9: DME 962–1213 MHz.	28
96	**C**	Figure 9: ATC 1030–1090 MHz.	28
97	**C**	Figure 9: LRRA 4,25–4,35 GHz.	28
98	**A**	Figure 9: Doppler Radar 8,8 GHz.	28
99	**B**	Figure 9: GPS L1: 1,5 GHz; L2: 1,2 GHz.	28
100	**C**	Figure (Radio Frequency Bands): HF – sky wave.	30
101	**B**	Figure (Radio Frequency Bands): VHF and UHF – space wave.	30
102	**A**	General rule of thumb is to multiply by 10 (3 MHz × 10 = 30).	21
103	**B**	Radar band table: X wavelength 5,8–2,5 cm.	27
104	**A**	Figure 9: 3 MHz ↔ 100 m and 30 MHz ↔ 10 m.	28
105	**A**	Figure (Radio Frequency Bands): Medium (MF) – ground/sky wave.	30

5 · Radio Wave Propagation: Ground Wave and Sky Wave

#	Ans	Why (LTT wording)	Page
106	**B**	Radio waves travel at approximately the speed of light.	32
107	**B**	Part of the wave moves outward in contact with the ground to form the GROUND WAVE.	32
108	**C**	The rest of the wave moves upward and outward to form the SKY WAVE.	32
109	**A**	Wave propagation defines the way in which waves are transmitted from an end to another.	34
110	**C**	The ground wave is responsible for most of the daytime broadcast reception.	36

#	Ans	Why (LTT wording)	Page
111	**C**	The sky wave is used for long-range high frequency daylight communication.	32
112	**A**	High frequencies (from HF and above) only travel in a direct line. (See conflict note.)	34
113	**C**	The ground wave is used for short-range communications at high frequencies with low power and for long-range communication at low frequencies with very high power.	32
114	**B**	Radio waves propagated into space are a radiant form of energy, similar to light and heat.	32
115	**C**	Space waves: VHF and higher travel in a Line of Sight (LOS).	34
116	**A**	Ground waves: MF and lower follow the curvature of the earth.	34
117	**B**	Attenuation = reduce/weaken; amplification = increase.	36
118	**A**	The conductivity of salt water is 5,000 times as great as that of dry soil.	36
119	**C**	Ground wave propagation usually lies in the range from kHz to a few MHz (generally up to 2 MHz).	37
120	**B**	The permissible frequency range for sky wave propagation lies between 3 MHz and 30 MHz.	39
121	**B**	The field pattern resembles a huge doughnut lying on the ground with the antenna in the hole at the centre.	32
122	**C**	Daytime reception from most commercial stations is carried by the ground wave.	32
123	**B**	At night, the sky wave provides a means for long-range contacts at somewhat lower frequencies.	32
124	**C**	The wave induces a voltage in the earth, setting up eddy currents; the energy is absorbed from the ground wave.	36
125	**A**	Increasing the frequency rapidly increases the attenuation, so ground wave transmission is limited to relatively low frequencies.	36
126	**B**	Shore-base transmitters use frequencies between 18 and 300 kHz with extremely high power.	36
127	**C**	The electrical properties of the earth along which the surface wave travels are relatively constant.	36
128	**B**	In localities with distinct rainy and dry seasons the moisture changes the conductivity of the soil.	36
129	**A**	Because of the superiority of surface wave conduction by salt water.	36
130	**A**	The waves oscillate parallel to the surface of the earth.	37
131	**C**	Some of the energy of the sky wave is refracted (bent) by the ionosphere so that it comes back toward the earth.	38
132	**B**	Some energy is lost in dissipation to particles of the atmospheric layers.	38
133	**A**	Advantages: large distance propagation, high frequency range, less atmospheric attenuation.	39
134	**B**	Disadvantage: long-distance propagation requires large-sized antennas.	39
135	**C**	Sky wave propagation is widely used in mobile and satellite communications. (See conflict note.)	39
136	**C**	Oscillations induce a wave of equal magnitude but opposite polarity on the surface.	37

#	Ans	Why (LTT wording)	Page
137	**B**	A receiver near the returning sky wave will receive strong signals even several hundred miles beyond the range of the ground wave.	38
138	**A**	Due to the presence of the ionosphere near and far during night and day respectively there is variation between day and night.	39

6 · Ionosphere and Ionospheric Effects

#	Ans	Why (LTT wording)	Page
139	**C**	The ionosphere is found in the rarefied atmosphere approximately 40 to 50 miles above the earth.	40
140	**A**	The "F" layer extends approximately from the 90 mile level to the upper limits of the ionosphere.	41
141	**B**	Densities of ionization at different heights make the ionosphere appear to have layers.	40
142	**A**	Its ionization is low and it has little effect on the propagation of radio waves except for the absorption of energy.	41
143	**B**	The "D" layer is present only during the day.	41
144	**C**	The maximum density of the "E" layer appears at about noon local time.	41
145	**C**	As a rule, the F2 layer is at its greatest density during early afternoon hours.	41
146	**A**	Shortly after sunset, the F1 and F2 layers recombine into a single F layer.	41
147	**C**	These patches are referred to as Sporadic-E ionisation.	41
148	**A**	Sporadic ionisations sometimes appear in considerable strength and actually prove harmful to electronic transmissions.	41
149	**B**	Some waves penetrate and pass entirely through it into space, never to return. Other waves penetrate but bend.	44
150	**B**	The angle made by the group 2 rays is called the CRITICAL ANGLE for that frequency.	44
151	**A**	Group 3 rays strike the ionosphere at the smallest angle that will be refracted and still return to the earth.	44
152	**B**	The highest frequency that can be sent directly upward and still be refracted back to the earth is called the CRITICAL FREQUENCY.	44
153	**B**	Figure 14: Skip Distance runs from A (antenna) to B (where the sky wave returns). Figure text.	46
154	**A**	This action is of great importance to daylight transmissions for distances up to 1,500 miles.	41
155	**C**	They are often present in sufficient number and intensity to enable good VHF transmissions over distances not normally possible.	41
156	**A**	It depends upon the angle at which the sky wave strikes the ionosphere, the frequency of the transmissions, and ion density.	44
157	**C**	If the frequency and angle are correct and the ionosphere is sufficiently dense, the wave will return to the earth.	44
158	**B**	The angle at which the group 1 rays strike the ionosphere is too nearly vertical for the rays to be returned to earth.	44
159	**B**	Any ray that leaves the antenna at an angle greater than this angle (θ) will penetrate the ionosphere.	44

#	Ans	Why (LTT wording)	Page
160	**A**	At any smaller angle the rays will be refracted but will not return to the earth.	44
161	**C**	It contains a much higher number of positive and negative ions.	40
162	**B**	At altitudes above 350 miles the particles of air are far too sparse.	40
163	**A**	The ionized atmosphere between 40 and 50 miles is called the "D" layer.	41
164	**C**	The band of atmosphere between 50 and 90 miles contains the "E" layer.	41
165	**B**	It is a well defined band with greatest density at an altitude of about 70 miles.	41
166	**C**	The negative ions are believed to be atoms raised to a high level by solar bombardment of ultra-violet and particle radiations.	40
167	**A**	The rotation of the earth, the annual course of the earth around the sun, and sun spots affect the number of ions.	40
168	**A**	It depends on the amount of air present, the strength of radiation from the sun, and the relation to the earth's magnetic field.	40
169	**B**	Below about 40 miles only a few ions are present because the rate of return to a normal energy level is high.	40
170	**A**	There is thought to be no sharp dividing line between layers; a demarcation is indicated for discussion.	40
171	**C**	Its presence greatly reduces the field intensity of transmissions that must pass through daylight zones.	41
172	**A**	This layer is strongest during daylight hours, and is also present, but much weaker, at night.	41
173	**C**	At night only one F layer is present; during the day it often separates into F1 and F2.	41
174	**B**	Figures 13: Night shows F and E; Day shows F2, F1, E and D.	43
175	**C**	The ionosphere acts as an electronic mirror and refracts (bends) the sky wave back to the earth.	44
176	**B**	The critical frequency varies with locality, time of day, season of the year and the sunspot cycle.	44
177	**B**	Nomograms and frequency tables predict the Maximum Usable Frequency (MUF) for every hour of the day for every locality.	45
178	**A**	They are prepared from data obtained experimentally from stations scattered all over the world.	45
179	**C**	Between the point where the ground wave is dissipated and the point where the first sky wave returns, no signals will be heard: the SKIP ZONE.	45
180	**C**	Signals reflected by the ionosphere return to the ground, are reflected again and return to the ionosphere.	45
181	**B**	The ionization of the E layer at midday is sometimes intense enough to refract frequencies up to 20 MHz back to earth.	41
182	**A**	As the frequency increases, the initial angle decreases.	44
183	**A**	At sufficiently high frequencies, regardless of the angle, the wave will not be returned to the earth.	44
184	**B**	The skip zone for the lower high frequencies (3 to 9 MHz) will be greater at night than during the day.	45

#	Ans	Why (LTT wording)	Page
185	**A**	Skip effects can be minimized by utilizing lower frequencies for nighttime communication.	45
186	**C**	As a general rule, as the frequency decreases, the skip distance decreases.	45
187	**A**	Multiple-hop may cause fading when transmissions with a different number of reflections superpose with different phases.	45
188	**B**	Figure 14: Critical angle decreases as frequency increases; refraction decreases as frequency increases.	47

7 · Antenna Fundamentals, Fields, Polarization and Radiation Pattern

#	Ans	Why (LTT wording)	Page
189	**C**	It therefore represents the link between transmitter and receiver.	48
190	**B**	Antennas are always supplied with an AC, typically at high frequencies.	52
191	**C**	With antennas the typical length is λ/4.	52
192	**B**	Every electromagnetic field consists of an electrical and a magnetic field with a 90 degree angle between them.	60
193	**A**	The two most common polarizations used for antennas are horizontal and vertical.	60
194	**C**	An antenna is a wave transformer for transporting line-bundled electromagnetic energy into free radiation energy (transmitter) or vice versa (receiver).	48
195	**A**	The current is greatest at the feed point; it is zero at the end of the wire.	50
196	**A**	The current distribution has the shape of a part of a sinus. As the current flows it creates a magnetic field.	50
197	**C**	Coils create a phase shift of 90° between voltage and current. The voltage leads the current.	54
198	**B**	With twin cables, the magnetic field H and electrical field E stay directly near the line, so hardly any radiation occurs.	54
199	**B**	By spreading the twin cable lines the current and voltage distributions keep their sine shape. The result is a dipole antenna.	56
200	**C**	This field is known as the antenna's short distance field.	58
201	**A**	Through an idea from Maxwell these traverse currents also produce a magnetic field.	58
202	**B**	The polarization of a radio wave is laid down as the direction of the electrical field.	60
203	**C**	In aviation: navigation is horizontal polarized; communication is vertical polarized.	60
204	**A**	Circular polarized waves are transmitted with helical antennas.	60
205	**A**	Circular polarization continuously alters its polarization.	60
206	**C**	A λ/2 dipole emits 90 degrees to its axis. In the direction of the axis, the emission is virtually zero.	64
207	**A**	This short distance field dissipates very fast and after a distance of 3–4 λ is practically no longer present.	58
208	**B**	The short distance field contains therefore reactive energy.	58
209	**B**	In the long distance field (r > 3–4 λ) E and H fields are in phase and active energy is transmitted.	58

#	Ans	Why (LTT wording)	Page
210	**C**	The sum of this vector decreases with the square of the distance and is dependent on the antenna's direction.	62
211	**C**	The radiation pattern shows a summary of the radiation distribution in different directions in space.	62
212	**A**	Transmission and reception antennae can electrically be dealt with the same way.	48
213	**A**	Mechanical differences are necessary due to varying high transmission power.	48
214	**C**	Important criteria: frequency range, band width, propagation characteristic, polarization.	48
215	**C**	An exact calculation is seldom possible, as surrounding influences can strongly influence the antenna size.	48
216	**B**	The strength of the field is determined by the current and its direction alters with the current.	52
217	**B**	The voltage is highest at the end of the wires but lowest at the feed point.	54
218	**C**	The electrical field is strongest at the end of the cable but zero at the feed point.	54
219	**A**	The current and voltage distributions do not change; they retain their sinus shape.	56
220	**A**	The field near the antenna ($r < 0{,}1\lambda$) is produced directly from the HF current and voltage.	58
221	**B**	In the short distance field H and E fields are 90° displaced.	58
222	**C**	The energy radiated consists of the energy of both the fields.	58
223	**A**	The propagation direction of the radiated energy is given as the Poynting vector S.	58
224	**C**	Figure: $S = E \times H$ in W (V·A) per cm^2.	59
225	**B**	A horizontally aligned dipole also transmits a horizontally polarized field.	60
226	**C**	With home use antennas the radio signal is horizontal polarized and the TV signal is vertical polarized.	60
227	**B**	Such a theoretical antenna is classified as an isotropic radiator, i.e. spherical radiator.	62
228	**C**	From a side view, the dipole emits with the shape of an eight.	64
229	**A**	If the eight rotates around the dipole, the shape of a donut is created.	64
230	**A**	With a top view, the dipole emits in any direction with the same intensity.	64
231	**B**	With reception antennas the characteristic is the same.	64
232	**C**	S is at right angles to the direction of the E and H fields ($S = E \times H$).	58

8 · Antenna Types

#	Ans	Why (LTT wording)	Page
233	**A**	A Yagi antenna consists of a dipole plus one or more directors.	67
234	**B**	These directors often are referred to as parasitic elements.	67
235	**C**	The rod antenna is also referred to as Marconi-Antenna.	69
236	**B**	It is used as a receiver antenna for long and medium wave reception.	71
237	**B**	Examples of surface radiators are the Horn Antenna and the Lens Antenna.	73
238	**A**	A horn antenna is in effect a funnel shaped waveguide.	73

#	Ans	Why (LTT wording)	Page
239	**C**	Free air impedance is 120π, approximately equal to 377 Ω.	73
240	**B**	Horn antennas are used mainly for frequencies of 1 GHz and above.	73
241	**B**	A radiating element (dipole or horn) is installed at the focal point of a rotational parabolic metallic reflector.	76
242	**A**	These are sometimes known as a dielectric lens.	79
243	**B**	Figure 29: Electrical Dipole and Magnetic Dipole.	82
244	**A**	The field of a dipole antenna is perfectly symmetric.	69
245	**A**	Because of its directional characteristic, it is especially suitable as a DF-Antenna.	71
246	**C**	It is **NOT** suitable as a transmitting antenna.	71
247	**C**	The funnel shape acts as a transformer, converting the impedance of the wave guide into the impedance of the free air.	73
248	**A**	To achieve high gain and bunching, large apertures, small flare angles and large overall lengths are necessary.	76
249	**A**	Low bunching in short horn antennas can be overcome by using lens and reflector antennas as phase correcting systems.	76
250	**C**	The focusing angle θ depends upon the applied frequency and the reflector diameter.	76
251	**B**	The slot antenna is the magnetic equivalent to the electrical dipole.	81
252	**B**	The superposition of the fields of both dipole and parasitic elements results in increased antenna gain in the direction of the antenna.	67
253	**C**	Parasitic elements shorter than the dipole act as a director while elements longer than the dipole act as a reflector.	67
254	**A**	The diagram of the field of a λ/4 rod antenna corresponds to one half of the field of a λ/2 dipole.	69
255	**C**	U0 depends on the loop area, the number of windings and the angle θ between the loop plane and the transmitter.	71
256	**C**	The more gradually the widening occurs and the greater the funnel length, the greater the gain.	73
257	**B**	By using a convex shaped dielectric lens with εr > 1, the beams at the centre and edge radiate in phase.	76
258	**A**	With the correct delay, all beams leave the lens at the same time, they are in parallel.	76
259	**A**	Example antenna types: horn antenna, slot antenna, Yagi-Uda antenna and rectangular patch antenna.	66
260	**B**	The loop antenna consists of one or more square or round shaped wire windings.	71
261	**B**	This antenna is also referred to as a magnetic dipole.	81
262	**C**	It is formed by a slot in a sufficiently sized metal sheet or by a slot in a waveguide.	81
263	**B**	The dipole excites the parasitic elements which then also start transmitting.	67
264	**B**	Due to the distance to the dipole a phase shift is created.	67
265	**C**	On the sides, the antenna gain is reduced as the fields cancel out each other at least partially.	67
266	**C**	The same effects apply if the Yagi Antenna is used for reception.	67

#	Ans	Why (LTT wording)	Page
267	**A**	Figure: spacing 0.15 λ.	68
268	**A**	In case the plane comes with a finite conductivity, the field would change accordingly.	69
269	**C**	The rod antenna is used wherever the mechanical dimensions must be as small as possible.	69
270	**B**	The loop can be seen as a coil in which a voltage is induced by the magnetic components of the field.	71
271	**A**	In surface radiators not only a thin wire but also a surface is involved in the radiations.	73
272	**C**	The flare angles depend on the height, width and length of the horn antenna.	74
273	**A**	The propagation velocity of an electromagnetic wave depends on εr.	76
274	**C**	The reflected waves are in phase in the aperture plane.	76
275	**B**	Dish: single direction, single beam. Lens: multiple direction, multiple beams.	78
276	**B**	Lens antennas use the principle of the converging lens to collimate incident divergent energy into plane waves.	79
277	**C**	Figure: Director λ/2 − 5 %, Reflector λ/2 + 5 %.	68
278	**A**	Figure: Reflector λ/2 + 5 %.	68
279	**A**	The field above such a plane would not change, even if the lower half of the dipole was removed.	69
280	**C**	The maximum voltage is induced when the loop plane matches the direction of propagation.	71
281	**B**	Favourable focusing results when the flare angle is less than 40° and L and A are large compared to λ.	73
282	**A**	Beams that hit the lens in the centre are delayed more than beams that hit the outer areas.	76
283	**B**	Due to the medium of the lens, the electrical path length is increased and the wave undergoes retardation.	79
284	**B**	The electrical field in the slot runs horizontally, so a vertical slot antenna produces a horizontally polarized wave.	81

9 · Antenna Characteristics and Antenna Tuning

#	Ans	Why (LTT wording)	Page
285	**C**	Lobe width is an indication of the directivity of an antenna.	83
286	**A**	The reference antenna is usually an isotropic antenna.	83
287	**B**	Normally, antenna gain is given in decibels.	83
288	**C**	The more the radiation is concentrated in a certain direction, the greater the field strength in this direction for a given transmitter power.	83
289	**C**	The increase in radiated power in a given direction compared to a reference antenna is called antenna gain G.	83
290	**B**	Antenna gain is a performance indicator gauged in comparison to a reference source.	85

#	Ans	Why (LTT wording)	Page
291	**A**	For this purpose mechanical tuning is possible; the antenna length is adjusted to the wave length.	86
292	**B**	The antenna tip can be L or T shaped or be fitted with an "umbrella".	86
293	**C**	Whenever an antenna is to be tuned, its length must be adjusted to the frequency either via mechanical means or electrically.	88
294	**A**	The locus is a diagram indicating the ohmic resistance and the reactance of the antenna over the frequency.	88
295	**A**	In order to tune the antenna to the frequency desired the reactance must be zero.	88
296	**B**	Figure: Frequency Range tuned by HF Coupler.	89
297	**B**	The resonant frequency is reduced, i.e. the antenna is electrically "lengthened".	86
298	**A**	On the X-axis the ohmic resistance is indicated, on the Y-axis the reactance.	88
299	**A**	It indicates the apex angle at which the radiated power has reduced to half of its maximum value.	83
300	**C**	At no point within the lobe width is the received power less than 0,5 PE max.	83
301	**A**	This value is equivalent to the 3 dB point with respect to the main direction.	83
302	**B**	The use of a directional antenna has the same effect as an increase in transmitter power.	83
303	**C**	In practice, λ/2-dipole antennas are often used as reference antennas.	83
304	**A**	Boresight is the axis of maximum gain / maximum radiated power.	85
305	**C**	Typically, a 30-degree beamwidth provides decent coverage at one metre away.	85
306	**C**	An isotropic antenna transmits/receives uniformly in all directions, exhibiting a gain of 0 dBi.	85
307	**A**	In free space the λ/2-dipole has an antenna gain of 1.64 compared to the isotropic antenna.	83
308	**B**	For long wave frequencies wavelengths can be several kilometres long, so the antenna length must be less than λ/4.	86
309	**C**	The conductors at the tips act as an end capacitance CE parallel to CA, resulting in a larger total capacitance.	86
310	**C**	f1: antenna too short, tuned by an inductor in series; the antenna is lengthened.	88
311	**B**	f2: antenna too long, shortened by a capacitor in series.	88
312	**A**	f3: antenna too short, lengthened by a capacitor in parallel.	88
313	**C**	f4: antenna too long, shortened by an inductor in parallel.	88

10 · RF Lines, Matching and Standing Waves

#	Ans	Why (LTT wording)	Page
314	**A**	Radio frequency lines are required for the transfer of RF energy with as little loss as possible.	90
315	**B**	Other applications: tuned resonant circuits, resistance transformation, matching between symmetric and asymmetric circuits.	90
316	**B**	An asymmetric line where a tubular conductor shields a concentric inner line is called co-axial cable.	90

#	Ans	Why (LTT wording)	Page
317	**A**	Typical impedances of symmetric twin cables start at approx. 150 Ω.	94
318	**C**	In electrical engineering this principle is known as power transfer theorem or Jacobi's Law.	96
319	**A**	Electrically: symmetric and asymmetric. Physically: shielded and unshielded.	90
320	**C**	In a symmetric cable both lines have equal capacity to ground (C10 = C20), and the RF signals have equal amplitude and opposite polarity.	90
321	**B**	The asymmetric cable has one line on RF potential and the other line grounded.	90
322	**A**	The electrical data of RF lines are determined by the physical construction and the type of insulation (dielectric) material.	92
323	**C**	Line inductance L' and capacity C' determine the ratio of ix and ux.	92
324	**B**	In general, the impedance of a cable is calculated as Z0 = √(L/C).	94
325	**C**	Their disadvantages are the higher dampening, lower voltage ratings and the lower maximum power that can be forwarded.	94
326	**A**	The impedances of transmitter and antenna must match.	96
327	**B**	In case an antenna with a non-matching impedance is connected to a RF line, a standing wave is created.	98
328	**B**	The SWR indicates how much the impedance of the antenna differs from the impedance of the line (or transmitter).	98
329	**A**	A ratio of 1 means that the impedances are identical; we have maximum power transfer.	98
330	**C**	With open or shortened lines the ratio is infinite.	98
331	**B**	Figure: Opposite Phase Reflection and In-Phase Reflection.	99
332	**B**	With a shorted or open ended line large standing waves for current and voltage appear.	102
333	**A**	One line capacity (C20) to ground is missing because ground itself is the conductor.	90
334	**C**	The inner wire has a capacity C12 against the shield, but no capacity C10 to ground.	90
335	**A**	With RF lines, a short or an open line create a hard reflection while smaller deviations create softer reflections.	98
336	**C**	With an open line there is no current flow at the end but the voltage reaches its maximum.	100
337	**C**	With a shortened line the maximum current flows at the end but the voltage is zero.	100
338	**B**	A shorted λ/4 (or open λ/2) line has frequency dependent resistive behaviour and behaves as a parallel resonant circuit.	102
339	**A**	With an open ended λ/4 line (or shorted λ/2 line) the line segment behaves as a series resonant circuit.	102
340	**C**	Lower impedances are possible with co-axial cables where 50–75 Ω are common.	94
341	**A**	The standard impedance of coaxial power cables in modern installations is 50 Ω.	94
342	**B**	They are easy to handle and can be produced at a reasonable price.	94

#	Ans	Why (LTT wording)	Page
343	**B**	Every line has an inductance L', a capacity C', a series resistance and a parallel resistance per length unit.	92
344	**A**	For most practical cases, the series resistance of the copper and the parallel resistance of the insulation can be ignored.	92
345	**B**	The voltage/current ratio of an infinitely long line determines the line impedance (Z0 = ux/ix).	94
346	**B**	Z0 depends on L and C, so it depends on the materials and the physical dimensions of the cable.	94
347	**A**	With HF, the lines must also have a matching impedance so that a constant impedance is found from transmitter to antenna.	96
348	**C**	Figure: matched when RL = Z0.	97
349	**C**	Hard materials like concrete reflect pretty well while soft materials like sand or clay absorb a part of the energy.	98
350	**A**	The reflected wave runs back and both waves superpose; they may amplify or cancel out.	98
351	**C**	The greater the ratio the greater the deviation between both the impedances.	98
352	**B**	SWR = Emax / Emin = Imax / Imin.	98
353	**C**	SWR = Zload / Z0 or Z0 / Zload, whichever is greater.	98
354	**A**	A very high voltage can create arcing and sparking in the dielectric, causing damage to the dielectric.	100
355	**B**	A very high current can create massively heated-up spots half a wavelength apart, causing burned spots or even small fires.	100
356	**C**	The hard reflection creates in-phase superposition of both waves in distances of half a wave length.	100
357	**A**	A shorted λ/4 line or an open λ/2 line behaves as a parallel resonant circuit.	102
358	**A**	The current standing wave at the input has the node and the voltage standing wave the loop.	102

11 · Resonant Circuits

#	Ans	Why (LTT wording)	Page
359	**B**	Mechanical oscillators consist of a mass which can be displaced against a return force (spring or the earth's gravitational force – pendulum).	105
360	**C**	Electric oscillator systems consist of inductors and capacitors.	105
361	**A**	Bandwidth B is the frequency range within which the resonance curve drops to a specified value on either side of the peak.	114
362	**B**	Resonant circuits are used for selective amplification, muting, or transformation of their frequencies.	116
363	**C**	At the inductor the current is stored by means of a magnetic field; the capacitor stores it in form of voltage.	105
364	**C**	At f0, XL = XC, UL equals UC; being 180° out of phase they cancel and a high current flows which is only limited by RS.	109
365	**A**	Bandwidth is determined by the quality factor Q.	114

#	Ans	Why (LTT wording)	Page
366	**B**	Selectivity is the ability of a resonant circuit to suppress other than the 1st resonant frequency, thus suppressing other radio stations.	114
367	**B**	The current imax at f0 causes voltages across L and C which can be much higher than the applied voltage U.	109
368	**C**	At f0, iL equals iC; being 180° out of phase they cancel each other, and only the current iR due to RP flows.	111
369	**B**	A series resonant circuit tuned to the unwanted signal is used as the ground leg in an antenna voltage divider.	116
370	**C**	A parallel resonant circuit tuned to the desired frequency is used as ground leg of an antenna voltage divider.	116
371	**B**	They are used for generation and selective amplification or damping of periodic processes.	105
372	**A**	A resistor is independent of frequency; its impedance is constant.	113
373	**A**	Inductors and capacitors create a time delay between voltage and current, so the energy travels between them.	105
374	**C**	The frequency is determined by the magnitude of the mass and the length l.	107
375	**C**	In a system without friction or other losses, their sum is constant at any instant.	107
376	**B**	Energy is introduced by an external voltage which sets up an electric field in the capacitor.	107
377	**B**	The process is repeated with a frequency determined by the dimensions of L and C.	107
378	**A**	The undamped oscillator frequency: f0 = 1 / (2π √(L C)).	107
379	**C**	Figure: electrical ↔ potential; magnetic ↔ kinetic.	108
380	**A**	Losses of L and C are shown combined in RS.	109
381	**A**	Their geometric sum is equal to the applied voltage U (2nd law of Kirchhoff).	109
382	**C**	Z is lowest at f0 because XL and XC cancel each other.	109
383	**B**	As RS is a real load, the phase angle between U and i is zero degrees.	109
384	**C**	Losses of L and C are represented by the parallel resistor Rp.	111
385	**A**	The current i is the geometric sum of the individual currents (1st law of Kirchhoff).	111
386	**B**	The apparent resistance Z equals RP at the resonant frequency.	111
387	**B**	The impedance of an inductor is proportional to the frequency.	113
388	**C**	For a capacitor its impedance decreases (though not linearly) as the frequency increases.	113
389	**A**	At the intersection, XL = XC; that frequency is called the resonant frequency fR.	113
390	**C**	Normally, this is −3 dB (1/√2 = 0.707) from the peak value 1 (0 dB).	114
391	**B**	Any other than the normal bandwidth of −3 dB must be specified.	114
392	**B**	Bandwidth B = f0 / Q.	114
393	**A**	The unwanted frequency is shorted to ground, but all other frequencies are unaffected.	116
394	**B**	Unwanted frequencies are shorted; only at the desired frequency the impedance and therefore the output voltage is high.	116

#	Ans	Why (LTT wording)	Page
395	**A**	The magnetic field reaches its maximum strength when the voltage is 0.	107
396	**A**	The magnetic field induces a voltage in the coil which charges the capacitor with reversed polarity.	107
397	**C**	Below f0, UC is greater: capacitive. Above f0, UL is greater: inductive.	109
398	**A**	Below f0, iL is greater: inductive. Above f0, iC is greater: capacitive.	111
399	**B**	In general a high selectivity is desirable but this is typically on the cost of the bandwidth.	114
400	**C**	Figure 44: Large Q factor makes the characteristic sharp; small Q makes it gradual.	115